# The Measurement Protocol Decides the Conclusion: A Methodological Study of CNN and Vision Transformer Adversarial Robustness Comparisons

Çağrı Şahin and Özal Yıldırım

***Abstract*—Comparisons of adversarial robustness across architecture families report conflicting conclusions, and the disagreement is usually attributed to models, training recipes, or datasets. We show that a substantial part of it comes from the measurement protocol itself. Using adversarially trained ResNet-18 and ViT-Tiny pairs on CIFAR-10, using three independent training runs per architecture, full-test-set evaluation with per-sample logging, and a fixed validation split held out before any training, we vary only how robustness and transfer are measured. Across four established conditioning protocols the measured CNN-to-ViT transfer asymmetry ranges from 4.4 to 14.6 points, a 3.3-fold spread, and the choice decides whether an equivalence test is accepted or rejected, although the direction of the effect is stable. Replicating the entire design on CIFAR-100 widens the per-seed spread from 10.5 to 13.6 points and preserves the sign in all twelve measurements, so the effect is not specific to one dataset. A further ablation that holds the training trajectory fixed and varies only the offline checkpoint-selection rule moves reported robust accuracy by 1.6 to 2.9 points, a fourth source of variance that is almost never reported. A third dataset sharpens this. On SVHN, where the same recipe leaves the two architectures only 1.85 points apart in clean accuracy, two of the four protocols report a CNN advantage and two report a ViT advantage, and the primary protocol accepts statistical equivalence: once the models are matched, the measurement choice decides not only the size of the reported asymmetry but its sign. A 3x3 matrix with a WideResNet-28-10 reference shows the deviation of unconditioned from conditioned rates is not merely correlated with the target's clean error but algebraically equal to it: given that clean-misclassified samples stay misclassified under attack (measured 0.989-1.000 over 36 directions), raw = clean error + conditioned rate x (1 - clean error) holds to within 0.41 points. Raw transfer rates are therefore contaminated by target weakness by construction, in any study that reports them. A conditional decomposition separates the CNN advantage into a clean-accuracy factor and a conditional-sensitivity factor; the attribution between the two changes between training runs even though the absolute ordering does not. Finally, three claims common in this literature fail to replicate under direct measurement: input gradients show no spatial localization difference, CLS attention entropy does not change under attack at n = 1000, and a ViT-specific transfer attack yields no transfer gain in this regime. Scale-invariant gradient sparsity differences and architecture-specific depth profiles of adversarial damage do survive. We close with reporting requirements for cross-architecture robustness studies.**

***Index Terms*—Adversarial robustness, convolutional neural networks, evaluation methodology, deep learning security, reproducibility, transfer attacks, Vision Transformer**

The authors are with the Department of Software Engineering, Fırat University, 23119 Elazığ, Turkey.
Corresponding author: Çağrı Şahin (e-mail: csahin@firat.edu.tr).

## I. Introduction

Deep neural networks are vulnerable to adversarial examples, carefully crafted perturbations that cause confident misclassification while remaining bounded by a small norm budget [1], [2]. As Vision Transformers (ViTs) [3], [4] joined convolutional neural networks (CNNs) [5] as a dominant architecture family, a natural question followed: which family is more robust, and why?

The literature answers this question inconsistently. Bai et al. [6] report that under matched training recipes transformers do not surpass CNNs, whereas Benz et al. [7] report ViTs to be more robust for standardly trained models; studies of cross-architecture transfer likewise disagree on whether attacks move more easily from CNNs to ViTs or the reverse [8]–[10]. These disagreements are usually attributed to differences in models, training recipes, or datasets, and the standard response is to compare under a matched recipe.

This paper makes a different observation. Even when the models, the data, the threat model, and the attack budget are all held fixed, the conclusion still depends on a choice that is rarely reported: the *measurement protocol*. Transfer studies must decide which samples enter the denominator (all test samples, those the target classifies correctly, those both models classify correctly, or those on which the source attack succeeded), and robustness comparisons must decide whether to report an absolute score or to separate it into clean accuracy and conditional susceptibility. These choices are usually treated as reporting details. We show that they are part of the result.

Our setting is deliberately narrow so that the measurement question is isolated. We adversarially train ResNet-18 and ViT-Tiny on CIFAR-10 [11] under a matched objective and threat model, with three independent training runs per architecture and a fixed validation split held out before *any* training stage, so that no model is selected on data it has seen. All evaluations run on the full test set with per-sample outcome logging, which makes paired statistics possible: the same samples can be re-weighted under different protocols without re-running an attack. We then vary the protocol, not the models, and measure how far the conclusion moves.

We also treat three claims that recur in architecture-comparison papers as measurable hypotheses rather than as interpretation: that CNN input gradients are spatially more localized, that adversarial perturbation degrades ViT attention, and that ViT-specific transfer attacks improve cross-architecture transfer. Each is tested directly with a purpose-built measurement.

Our contributions are as follows. We first quantify protocol sensitivity. On identical models and data, the measured cross-architecture transfer asymmetry ranges from $+4.4$ to $+14.6$ points across four established conditioning protocols, a 3.3-fold spread, whereas retraining both architectures from scratch moves the same quantity by under $1.5$ points. The protocol decides whether an equivalence test is accepted or rejected, while the direction of the asymmetry, unlike its magnitude, is stable. Repeating the full design on CIFAR-100 widens the mean per-seed protocol spread to $13.58 \pm 1.71$ points and preserves the sign in all twelve measurements. Holding a training trajectory fixed and varying only the offline checkpoint-selection rule moves reported PGD-10 accuracy by $1.58$ to $2.85$ points, which places selection alongside seeds, attacks, and protocols as a reportable source of variance.

We then show that unconditioned transfer rates are confounded, and measure by how much. In a $3 \times 3$ transfer matrix that adds a WideResNet-28-10 reference, the deviation of raw from conditioned rates is not an empirical correlation but an identity. Because samples the target already misclassifies are essentially never rescued by the perturbation ($0.989$ to $1.000$ over $36$ directions), the raw rate equals the clean error plus the conditioned rate scaled by one minus that error, to within $0.41$ points. Incoming transfer also tracks the target's own vulnerability rather than the source architecture ($r = 0.986$ across three targets), which reframes the asymmetry as a property of the weaker target rather than of stronger CNN-crafted attacks. A conditional decomposition completes the picture: absolute robustness factors exactly into clean accuracy and conditional susceptibility, and reporting both matters, because the attribution between them changed between our earlier single-run checkpoints and three fresh runs while the absolute ordering did not.

We report three negative results. Scale-invariant sparsity differences do not correspond to spatial localization differences, on four spatial measures of which three are null; CLS attention entropy does not change measurably under attack at $n = 1000$; and Token Gradient Regularization [12], a ViT-specific transfer attack, gives no transfer gain in this regime, losing to plain MI-FGSM under a matched budget. Three findings do survive. Gradient sparsity differences reproduce across independent checkpoint sets, are created by adversarial training rather than by architecture, and are accompanied by an alignment difference that exists only in absolute cosine. Adversarial damage accumulates at qualitatively different depths in the two architectures, and the measured drift profile depends on the token aggregation used to compute it.

We close with a short set of reporting requirements that follow from these measurements.

Section II reviews adversarial evaluation practice and architecture-comparison studies. Section III describes the training protocol, the conditioning protocols, and the measurements. Section IV reports results. Section V discusses implications and limitations, and Section VI concludes.

## II. Related Work

### A. Adversarial Attacks

Adversarial attacks aim to find minimal perturbations that cause misclassification. The Fast Gradient Sign Method (FGSM) [2] generates perturbations in a single step by following the sign of the loss gradient, with the perturbation magnitude bounded under the $L_\infty$ norm. Projected Gradient Descent (PGD) [13] extends FGSM through iterative optimization, taking multiple gradient-sign steps and projecting back onto the $\epsilon$-ball after each step (formal definitions in Section III-C).

The Carlini & Wagner (C&W) attack [14] formulates adversarial example generation as an optimization problem, often achieving higher success rates but at increased computational cost. More recently, AutoAttack [15] combines multiple attack strategies (APGD-CE, APGD-T, FAB-T, Square Attack) to provide a reliable robustness evaluation, becoming the de facto standard for benchmarking.

### B. Defense Mechanisms

Adversarial training (AT) [13] remains the most effective defense, training models on adversarial examples generated during each batch:

$$\min_\theta \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[ \max_{\|\delta\|_\infty \le \epsilon} \mathcal{L}(f_\theta(x+\delta), y) \right] \tag{1}$$

TRADES [16] provides a theoretically-grounded trade-off between natural accuracy and robustness:

$$\mathcal{L}_{\text{TRADES}} = \mathcal{L}_{CE}(f(x), y) + \beta \cdot \text{KL}(f(x) \| f(x_{adv})) \tag{2}$$

where $\beta$ controls the trade-off parameter.

MART [17] addresses the issue of misclassified examples by emphasizing them during training. Other approaches include input transformation [18], certified defenses via randomized smoothing [19], and ensemble methods [20]. Recent advances in certified robustness include Adaptive Randomized Smoothing [21], which improves certification radii through multi-step defenses.

### C. CNN Robustness

Extensive research has characterized the adversarial robustness of CNN architectures. Madry et al. [13] demonstrated that adversarial training with PGD attacks provides strong empirical robustness. The RobustBench leaderboard [22] tracks state-of-the-art CNN robustness, with WideResNet variants achieving the highest robust accuracy on CIFAR-10.

Studies have shown that model capacity correlates with achievable robustness [13], with wider and deeper networks demonstrating improved robust accuracy. Architecture-specific findings include the role of skip connections in adversarial transferability [23] and the distinct batch-normalization statistics that adversarial training induces at scale [24]; relatedly, auxiliary batch-normalization branches allow adversarial examples to improve clean accuracy [25]. Adversarial training is also known to reshape input-gradient structure itself, yielding more perceptually aligned [26] and sparser [27] gradients.

### D. Vision Transformer Robustness

Vision Transformers [3], building on the self-attention mechanism introduced by Vaswani et al. [4], process images as sequences of patches; their internal representations differ measurably from CNNs' layer by layer [28]. Initial studies suggested that ViTs exhibit some inherent robustness properties due to their global receptive field [29]. Bai et al. [6] systematically investigated whether transformers are more robust than CNNs, finding that under fair training recipes transformers are not more adversarially robust than CNNs, while exhibiting superior robustness under distribution shift attributed to self-attention; Benz et al. [7] reported an early direct comparison including MLP-Mixer and linked the differences to frequency preferences. Our work extends this line of inquiry by analyzing adversarially trained models of comparable scale and providing behavioral characterizations through gradient characteristics and feature degradation analysis. Subsequent work revealed that ViTs remain vulnerable to adversarial attacks, particularly gradient-based methods [8], with patch-targeted attacks exploiting the attention mechanism directly [30] and attention shown to shift measurably under patch perturbations [31].

Shao et al. [32] demonstrated that adversarial training can improve ViT robustness and that ViT features rely less on high-frequency content, though challenges remain due to training instability. ViT robustness to common corruptions and occlusions has also been characterized [33]. More recently, Jain et al. [34] proposed Adaptive Attention Scaling Adversarial Training (AAS-AT) to address ViT-specific training challenges, achieving state-of-the-art robustness on CIFAR-10 and ImageNet-100; with diffusion-augmented adversarial training, ViTs have even been reported to surpass WideResNets on small-scale datasets [35]. Zhu et al. [36] strengthened transferable attacks on ViTs through gradient normalization scaling and high-frequency adaptation. Despite these advances, comprehensive behavioral comparisons between CNN and ViT robustness under unified evaluation frameworks remain limited.

### E. Transfer Attacks

Adversarial examples often transfer between models, enabling black-box attacks [37]. The transferability depends on factors including model architecture, training data, and attack methods [38]. Foundational transfer studies already evaluate transferability on correctly classified samples [39], and standard attack papers condition their success rates for the same reason [40]; recent evaluation guidelines make such protocol choices explicit [41]. In ensemble design, Ravikumar et al. [9] restrict transfer measurement to samples correctly classified by both models precisely to account for accuracy differences between them, the same confound our conditioned metric controls.

Cross-architecture transfer between CNNs and ViTs has received increasing attention. Mahmood et al. [8] measured cross-architecture transfer with a both-correct protocol and found notably low CNN$\leftrightarrow$ViT transferability; Naseer et al. [10] showed that adversarial examples crafted *on* ViTs transfer poorly under conventional attacks and proposed self-ensemble and token refinement to improve ViT-sourced transferability, and Wei et al. [42] improved ViT-sourced transfer by bypassing attention gradients and patch-wise dropout. Building on these, Token Gradient Regularization (TGR) [12] crafts transferable adversarial samples by exploiting ViT's attention components, while Momentum Integrated Gradients (MIG) [43] achieves improved transferability across both ViTs and CNNs. Chen et al. [44] systematically analyze the factors affecting cross-architecture transferability, finding that architectural inductive biases, with CNNs preferring high-frequency information while ViTs favor low-frequency patterns, significantly impact transfer success. Understanding these transfer patterns is crucial for developing robust ensemble systems and evaluating real-world attack scenarios.

### F. Research Gap

While individual studies have examined CNN and ViT robustness separately, and the pitfalls of unfair CNN and transformer comparisons have been highlighted before [6], [45], existing comparative work typically reports robustness scores in isolation, often with inconsistent perturbation budgets, attack configurations, or model scales, rather than pairing the comparison with behavioral analysis. A recent study in this journal compares ViT and CNN adversarial robustness directly [46]; our work differs by evaluating on the full test set with AutoAttack and paired statistics, by controlling transfer measurement for clean-accuracy differences, and by accompanying the comparison with gradient- and feature-level behavioral analyses.

Our work engages the measurement question directly and makes it the object of study. Conditioning transfer evaluation on correctly classified samples is established practice [8], [9], [39], [40], and individual papers pick one protocol and proceed. What is missing is a controlled estimate of how much that pick matters. We supply it by holding the models, the data, the threat model, and the attack budget fixed and varying only the protocol, and we find that the measured asymmetry moves by $10.45 \pm 0.76$ points across the four protocols, a 3.3-fold spread, while retraining both architectures moves that same quantity by under $1.5$ points. We further show that the confound in unconditioned rates is not a tendency but an identity ($r_{\text{raw}} - r_{\text{cond}} = e\,(1 - r_{\text{cond}})$, verified to within $0.41$ points over 36 directions), which turns a known qualitative caveat into a correction that can be applied directly to published results.

Three further claims recur in this literature that our design lets us test rather than assume: that CNN gradients are spatially more localized, that adversarial perturbation degrades ViT attention, and that transformer-specific transfer attacks [10], [12] improve cross-architecture transfer. We measure each directly, and none holds in our setting, and we report these results because an untested assumption of this kind is exactly what a protocol-sensitive comparison cannot afford.

## III. Methodology

### A. Threat Model

We consider the standard $L_\infty$ threat model where an adversary can perturb input images within a bounded region:

$$\mathcal{B}_\epsilon(x) = \{x' : \|x' - x\|_\infty \leq \epsilon\} \tag{3}$$

Following established conventions for CIFAR-10 [11], [22], we use $\epsilon = 8/255 \approx 0.031$ as the primary perturbation budget, with additional full-test PGD-10 evaluations at $\epsilon \in \{2/255, 4/255, 16/255\}$ for the epsilon sweep analysis.

We evaluate under both white-box (full model access) and black-box (transfer attack) scenarios. For white-box attacks, the adversary has complete knowledge of model architecture and parameters. For transfer attacks, adversarial examples are generated on a source model and evaluated on a different target model.

### B. Model Architectures

*a) CNN Models:* **ResNet-18** [5]: A widely-used baseline CNN with residual connections, containing 11.2M parameters. We use the standard architecture designed for CIFAR-10 (initial 3×3 convolution instead of 7×7).

**WideResNet-28-10** [47]: A wider variant of ResNet with 36.5M parameters, representing the state-of-the-art architecture for adversarial robustness on CIFAR-10. We use the pretrained robust checkpoint of Gowal et al. [48] distributed via RobustBench [22]; note that this model was trained with additional unlabeled data and a stronger recipe than our standard adversarial training, so it serves as a reference point rather than a matched comparison.

*b) Vision Transformer Models:* **ViT-Tiny**: We use the ViT-Tiny architecture from the `timm` library [49], adapted for CIFAR-10 by bilinearly upsampling inputs to 224×224. The architecture uses 16×16 patches (196 patches per image), 192-dimensional embeddings, 12 transformer blocks with 3 attention heads each, MLP ratio of 4, and totals 5.7M parameters. The model is trained *from scratch* on CIFAR-10 (no ImageNet pretraining) with basic augmentation (random crop and horizontal flip), then adversarially fine-tuned. Since ViT robustness on small datasets is known to be sensitive to the training recipe [50], [51], our architecture-level conclusions should be interpreted as holding *under this baseline training recipe* rather than for optimally trained ViTs.

### C. Attack Methods

*a) FGSM:* Single-step attack with step size equal to the perturbation budget:

$$x_{adv} = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}_{CE}(f(x), y)) \tag{4}$$

*b) PGD:* Iterative attack with step size $\alpha = 2/255$ and random initialization:

$$x^{t+1} = \Pi_{\mathcal{B}_\epsilon(x)}\left(x^t + \alpha \cdot \text{sign}(\nabla_{x^t}\mathcal{L}_{CE}(f(x^t), y))\right) \tag{5}$$

with $x^0 = x + \mathcal{U}(-\epsilon, \epsilon)$. We use PGD-10 (10 iterations) for both adversarial training and evaluation throughout this work (Table I).

*c) AutoAttack:* We use AutoAttack [15], the standardized ensemble attack combining APGD-CE (Auto-PGD with cross-entropy loss), APGD-T (Auto-PGD with targeted DLR loss), FAB-T (targeted Fast Adaptive Boundary attack), and Square Attack (score-based black-box attack). This ensemble is a standardized strong benchmark for robustness evaluation.

### D. Defense Methods

*a) Adversarial Training (AT):* We employ a mixed adversarial training objective that combines clean and adversarial losses:

$$\mathcal{L} = (1-\lambda)\,\mathcal{L}_{CE}(f_\theta(x), y) + \lambda\,\mathcal{L}_{CE}(f_\theta(x_{adv}), y) \tag{6}$$

where $\lambda = 0.5$ balances standard classification performance with adversarial robustness. The inner maximization generates adversarial examples using PGD-10 ($\alpha = 2/255$, $\epsilon = 8/255$), computed with the model in training mode so that attack generation and the parameter update see consistent batch-normalization statistics, following standard adversarial-training implementations [52]. This mixed objective, widely adopted in practice, helps maintain clean accuracy while building robustness, as training solely on adversarial examples can degrade natural performance. While alternative formulations such as TRADES [16] offer theoretically grounded trade-off mechanisms, this work focuses on standard adversarial training to isolate architectural effects from defense-specific interactions.

### E. Training Protocol

*a) Clean Training:* For CNNs, we use SGD optimizer with initial learning rate 0.1, cosine annealing schedule [53], weight decay $5 \times 10^{-4}$, and 200 epochs. For ViTs, we use AdamW optimizer [54] with initial learning rate $10^{-3}$, cosine annealing, weight decay 0.05, and 200 epochs.

*b) Adversarial Training:* Starting from our clean-trained checkpoints, we fine-tune with reduced learning rates: CNNs use SGD with LR $10^{-3}$ and cosine annealing for up to 100 epochs, while ViTs use AdamW with the same learning rate and schedule. Gradients are clipped to a global $L_2$ norm of 10, and batches with non-finite loss are skipped. Data augmentation includes random cropping with 4-pixel padding and horizontal flipping; all models consume raw pixel inputs in $[0, 1]$ without mean/std normalization, and attacks and epsilon budgets operate directly in this pixel space. The training batch size is 128 for the CNN and 64 for the ViT (evaluation batch sizes vary by analysis). Early stopping uses patience 20 with a 0.1 percentage-point improvement threshold.

*c) Validation Split and Repetitions:* Model selection and early stopping use PGD-10 adversarial accuracy on a fixed 2,000-sample validation split. The split is drawn once (seed 777, indices stored with the code) and is shared by every model in the study, and it is excluded from the training set of *both* the clean and the adversarial stage, so no checkpoint is ever selected on data it was trained on. The test set is touched only by the final post-hoc evaluations. We repeat the full pipeline, clean pre-training followed by adversarial fine-tuning, three times per architecture with independent seeds (1001 to 1003 for ResNet-18, 2001 to 2003 for ViT-Tiny), and pair the runs by index so that every cross-architecture comparison is between models trained under the same repetition. All tables report the mean and standard deviation over these three runs.

For one comparison in Section IV-A we also refer to an earlier pair of checkpoints trained before this protocol was adopted, whose validation split had been part of clean pre-training. We report them explicitly as a contrast, not as results, because the difference between the two checkpoint sets is itself evidence about how far a single-run conclusion can move.

### F. Evaluation Metrics

*a) Robustness Metrics:* We evaluate models using three standard metrics: clean accuracy (classification performance on unperturbed test images), robust accuracy (classification performance on adversarial examples), and attack success rate (fraction of correctly classified clean images that become misclassified after attack). For white-box evaluations we additionally report the *conditioned fooling rate*, the fraction of clean-correct samples that the attack flips, and its complement, conditioned survival. Because both PGD and AutoAttack count clean-misclassified samples as non-robust in our per-sample logs, absolute robust accuracy factorizes exactly as robust $=$ clean $\times$ conditioned survival, which lets us decompose robustness differences into a clean-accuracy component and a conditional-susceptibility component. Finally, we report accuracies on the *both-correct* subset (samples that both compared models classify correctly on clean inputs), which yields an exactly paired comparison on a shared sample set.

*b) Transfer Attack Metrics:* Our primary transfer metric is the *conditioned fooling rate*: among the samples that the target model classifies correctly on clean inputs, the fraction that becomes misclassified under adversarial examples crafted on the source model. Given adversarial examples $\{x^S_{adv}\}$ generated against source model $f_S$, the conditioned transfer rate to target model $f_T$ is:

$$T_c(f_S \to f_T) = \frac{\sum_{i=1}^n \mathbf{1}[f_T(x_i) = y_i]\,\mathbf{1}[f_T(x^S_{adv,i}) \neq y_i]}{\sum_{i=1}^n \mathbf{1}[f_T(x_i) = y_i]} \tag{7}$$

Conditioning transfer evaluation on correctly classified samples follows established practice [8], [9], [39]–[41] and is essential in cross-architecture comparisons: the raw misclassification rate $\frac{1}{n}\sum_i \mathbf{1}[f_T(x^S_{adv,i}) \neq y_i]$ includes the target's clean error, so a target with lower clean accuracy appears more "vulnerable" to transfer even when no additional samples are flipped by the attack. We report the raw rate as a secondary metric for transparency and to quantify the size of this confound. Because the conditioning set itself is a protocol choice, we additionally report two established alternatives: the *both-correct* protocol, which conditions on samples that both the source and the target classify correctly [8], [9], and the *successful-source* protocol, which conditions on target-correct samples whose attack succeeded on the source in the white-box sense. As Section IV-B shows, these protocols can yield qualitatively different asymmetry conclusions, so we treat the protocol as part of the reported result rather than an implementation detail. Transfer rates are computed on the full test set (10,000 samples) with per-sample outcomes logged, and we report 95% confidence intervals obtained by percentile bootstrap over samples (10,000 resamples).

*c) Gradient Analysis Metrics:* We analyze gradient properties through three complementary metric families, computing gradients of per-sample losses so that magnitudes are independent of the evaluation batch size. Gradient norm ($\|\nabla_x \mathcal{L}\|_2$ and $\|\nabla_x \mathcal{L}\|_\infty$) measures overall sensitivity magnitude. Gradient sparsity is quantified with scale-invariant measures: Hoyer sparsity, the Gini coefficient of $|g|$, and the fraction of components below 1% of the per-sample maximum, since fixed absolute thresholds conflate gradient scale with structure; Hoyer and Gini satisfy the scale-invariance axioms formalized by Hurley and Rickard [55]. Gradient alignment measures the average pairwise absolute cosine similarity across samples (computed over all pairs of the 500 analysis samples):

$$\text{Alignment} = \frac{1}{N(N-1)} \sum_{i \neq j} \left| \frac{\nabla_{x_i}\mathcal{L} \cdot \nabla_{x_j}\mathcal{L}}{\|\nabla_{x_i}\mathcal{L}\| \|\nabla_{x_j}\mathcal{L}\|} \right| \tag{8}$$

We use the absolute cosine because shared sensitivity *axes* matter for attack geometry regardless of sign; since sign coherence is what universal-perturbation arguments additionally require, we also report the signed mean cosine, which turns out to be near zero for every model (Section IV-F). High absolute alignment indicates similar perturbation axes across inputs.

*d) Spatial Locality Metrics:* Sparsity measures how many gradient components are large but says nothing about where they lie in the image, so we measure spatial structure separately on the channel-summed energy map $e = \sum_c (\nabla_x \mathcal{L})_c^2$, normalized to sum to one. We report the smallest pixel fraction whose energy sums to 50% and to 90% (smaller means more localized), the Shannon entropy of $e$ viewed as a distribution over pixels (lower means more localized), and Moran's $I$ with 4-neighbourhood weights, which is positive when energy clusters in adjacent pixels. All four are invariant to the scale of the gradient, and all are computed per sample so that the two architectures can be compared with paired tests on identical inputs.

*e) Token Aggregation for Layer-wise Analysis:* A transformer block emits one vector per token, so summarizing a block requires a choice. We report three: the CLS token alone, the mean over the 196 patch tokens, and the concatenation of all tokens flattened per sample. For the CNN each residual block output is flattened over its spatial dimensions. We report all three for the ViT because, as Section IV-G shows, they give different drift magnitudes and place the minimum at different depths.

*f) Attention Metrics:* For the ViT we extract the post-softmax CLS-to-patch attention row of each block, averaged over heads. We report its Shannon entropy, which measures how broadly attention is spread, and its displacement under attack, measured as the total variation distance between the clean and adversarial attention distributions (bounded by 1). Both are computed per sample and averaged, on $n = 1000$ samples per run.

*g) ViT-Specific Transfer Attack:* To test whether the transfer results depend on using architecture-agnostic attacks, we additionally generate source examples with Token Gradient

Regularization (TGR) [12], which zeroes the extreme-valued entries of the token-gradient tensors in the attention, QKV, and MLP paths during backpropagation and scales the remaining gradient by $\gamma = 0.5$. Our implementation adapts the method to our CIFAR-10 ViT wrapper and to the budget used throughout this paper ($\epsilon = 8/255$, $\alpha = 2/255$, 10 steps, momentum 1.0) rather than the original ImageNet defaults; we therefore report it as an adaptation. Plain MI-FGSM [40] examples are generated in the same run on the same samples, giving a paired control at an identical budget.

*h) Feature Degradation Metrics:* To analyze how adversarial perturbations affect ViT representations across layers, we compute three metrics at each layer $\ell$. Feature cosine similarity measures directional preservation of the representation:

$$\text{Sim}_\ell = \frac{h_\ell(x) \cdot h_\ell(x_{adv})}{\|h_\ell(x)\| \|h_\ell(x_{adv})\|} \tag{9}$$

where $h_\ell$ denotes the representation at layer $\ell$. Feature norm change captures magnitude effects:

$$\Delta_\ell = \frac{\|h_\ell(x_{adv})\| - \|h_\ell(x)\|}{\|h_\ell(x)\|} \times 100\% \tag{10}$$

Feature $L_2$ distance ($\|h_\ell(x) - h_\ell(x_{adv})\|_2$) quantifies absolute deviation.

### G. Statistical Validation

We separate four sources of variance and report each where it applies. *Training-run variance* is the one that bears on architecture comparisons: every table reports the mean and standard deviation over three independent runs per architecture (Section III-E). *Attack-initialization variance* is measured by re-running PGD-10 with three attack seeds (42, 123, 456) on a fixed subset; the evaluated samples and their order are fixed, so the seeds vary only the random start of the attack. *Protocol variance* is the spread across conditioning protocols, which we treat as a result rather than as noise (Section IV-B). *Checkpoint-selection variance* is the spread produced by the offline rule that picks the final checkpoint from a fixed trajectory, namely validation split, early-stopping patience, and curve smoothing, measured over eighteen combinations while the trajectory itself is held constant. Section IV-I compares their magnitudes.

Paired tests are used wherever the same samples are scored under two conditions: McNemar's exact test for paired binary outcomes, Wilcoxon signed-rank with Holm correction for the three sparsity measures, paired bootstrap confidence intervals (10,000 resamples) for differences, and a sign-flip permutation test (20,000 permutations) for the both-correct transfer difference. Equivalence claims use two one-sided tests (TOST) with the margin stated at each use. All evaluation and analysis scripts fix random seeds (seed 42, including CUDA determinism flags), so every reported number is reproducible from the released pipeline.

## IV. Experiments

### A. Robustness Comparison

Table I presents the main robustness comparison across all evaluated models. Every cell for our models is the mean over three independent training runs per architecture, with the standard deviation over those runs; each run is evaluated on the full 10,000-sample test set.

TABLE I: Main robustness results on CIFAR-10 ($\epsilon = 8/255$), mean$\pm$std over three independent training runs per architecture. AA denotes AutoAttack with the standard suite.

| Model | Defense | Params | Clean | FGSM | PGD-10 | AA |
|---|---|---|---|---|---|---|
| *CNN Models* | | | | | | |
| ResNet-18 | None | 11.2M | 95.25±0.15 | 40.29±1.77 | 0.03±0.03 | – |
| ResNet-18 | AT | 11.2M | 85.78±0.36 | 53.46±0.08 | 44.11±0.50 | **37.93±0.14** |
| WRN-28-10 | AT† | 36.5M | 89.48 | 70.91 | 66.92 | 62.76 |
| *Vision Transformer Models* | | | | | | |
| ViT-Tiny | None | 5.7M | 80.09±0.60 | 6.10±1.47 | 0.05±0.05 | – |
| ViT-Tiny | AT | 5.7M | 73.53±0.55 | 36.31±0.42 | 32.69±0.22 | **29.14±0.40** |

†Robust checkpoint of Gowal et al. [48] via RobustBench [22]; a single external checkpoint, so no standard deviation applies. Clean/FGSM/PGD-10: our local full-test evaluations; AA: the RobustBench-reported figure (we did not rerun AutoAttack on it).

TABLE II: Conditional decomposition of white-box robustness on the full test set, mean$\pm$std over three runs. The lower block repeats the same quantities on our earlier single-run checkpoints.

| Model | Cond. fooling (%) PGD-10 | Cond. fooling (%) AA | Both-correct robust (%) PGD-10 | Both-correct robust (%) AA |
|---|---|---|---|---|
| ResNet-18 AT | **48.58**±0.80 | **55.78**±0.23 | **59.86**±0.35 | **51.89**±0.49 |
| ViT-Tiny AT | 55.53±0.50 | 60.37±0.33 | 46.11±0.59 | 41.15±0.39 |
| *Earlier single run, validation split seen during clean pre-training (n=7,260)* | | | | |
| ResNet-18 AT | 52.15 | 58.16 | **54.92** | **48.15** |
| ViT-Tiny AT | 52.33 | **56.46** | 49.61 | 45.33 |

The CNN holds an *absolute* robustness advantage under every attack: ResNet-18 AT reaches $37.93 \pm 0.14\%$ robust accuracy under AutoAttack against $29.14 \pm 0.40\%$ for ViT-Tiny AT. Because both models of a pair are evaluated on the same 10,000 samples with per-sample outcomes logged, McNemar's paired test applies within each run; it favors the CNN decisively in all three pairs ($p < 10^{-84}$ under AutoAttack, $p < 10^{-120}$ under PGD-10). Seed-level variation is small throughout for the adversarially trained pair, at most 0.55 points on any reported quantity, so the ordering is not a product of run-to-run noise; the undefended baselines vary more under FGSM (up to 1.8 points), but no claim rests on them. The CNN also retains higher clean accuracy (85.78 vs 73.53%, a 12.3-point gap), reflecting the difficulty of training ViTs on small datasets from scratch; Table II separates that contribution from conditional susceptibility. AutoAttack robustness is consistently below PGD-10 robustness for both architectures (6.2 points for ResNet-18, 3.6 for ViT-Tiny), confirming the value of the stronger ensemble. The WideResNet-28-10 reference [48], at 62.76%, shows how far capacity, additional training data, and a stronger recipe jointly push robustness beyond the architectural choice; we keep it visually and analytically separate from the matched pair throughout.

Table II decomposes the absolute scores into their clean-accuracy and conditional-susceptibility components (Section III-F). Because misclassified clean inputs count as non-robust, the factorization is exact: for AutoAttack it reads $85.78 \times 0.442 = 37.93$ for the CNN and $73.53 \times 0.396 = 29.14$

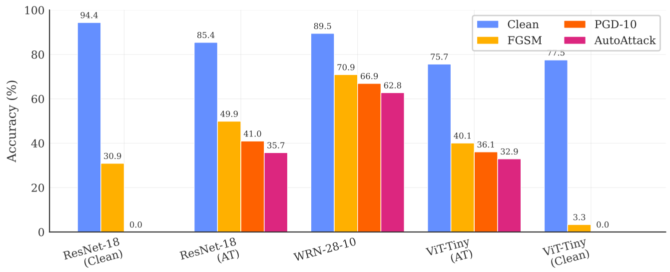



Fig. 1: Clean, FGSM, PGD-10, and AutoAttack accuracy at $\epsilon = 8/255$ on the full CIFAR-10 test set. The WideResNet AutoAttack bar is the value reported by RobustBench.

for the ViT. Both factors favor the CNN. It starts from a 12.3-point higher clean accuracy *and* loses a smaller fraction of what it solves, with conditional fooling of 48.58 versus 55.53% under PGD-10 and 55.78 versus 60.37% under AutoAttack. On the jointly solved subset, the $n = 7{,}061 \pm 68$ samples that both models of a pair classify correctly on clean inputs and therefore an exactly paired comparison on shared inputs, the CNN leads by 13.8 and 10.7 points (McNemar $p < 10^{-84}$ in every run).

The decomposition is worth reporting even when the two factors agree, because they need not, and because their relative weight is less stable than the ordering they produce. Our first checkpoints, selected with a validation split that clean pre-training had already seen, gave the same absolute ordering but the opposite conditional one: fooling rates that were indistinguishable under PGD-10 (52.15 versus 52.33%) and slightly ViT-favorable under AutoAttack (58.16 versus 56.46%), which supported the reading that the CNN advantage was entirely a clean-accuracy effect (lower block of Table II). None of the three corrected runs reproduces that reading, and the three agree with one another far more closely than any of them agrees with the earlier run. We cannot isolate a single cause, since the corrected protocol changed the validation handling and the training run at the same time; notably, a simple data-quantity explanation does not hold, because the corrected clean models are *better* than the earlier ones despite seeing 2,000 fewer images ($80.09 \pm 0.60$ versus 77.50% for the ViT). The defensible conclusion is narrower and more useful than a mechanism: an attribution of this kind should not be drawn from a single training run, and a paper that reports only the absolute score hides which factor is doing the work.

Figure 1 visualizes this robustness gap across architectures. To further understand the effect of perturbation magnitude, Figure 2 shows PGD-10 robust accuracy across perturbation budgets $\epsilon \in \{2/255, 4/255, 8/255, 16/255\}$ on the full test set.

Figure 3 provides visual examples of adversarial perturbations and their effects on model predictions.

### B. Transfer Attack Analysis

This section contains the paper's central measurement. We craft PGD-10 adversarial examples on a source model and evaluate them on a target, on the full test set, for each of the

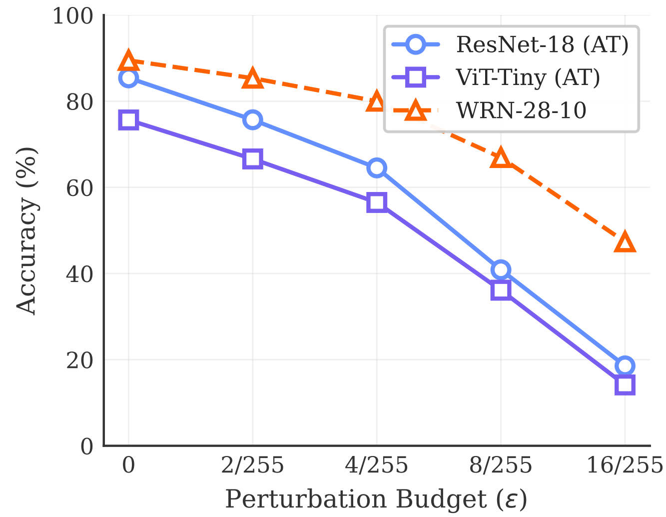



Fig. 2: PGD-10 robust accuracy vs. perturbation budget ($\epsilon$), measured on the full test set. Both AT architectures degrade with similar shapes as epsilon increases, with the CNN maintaining a consistent margin.

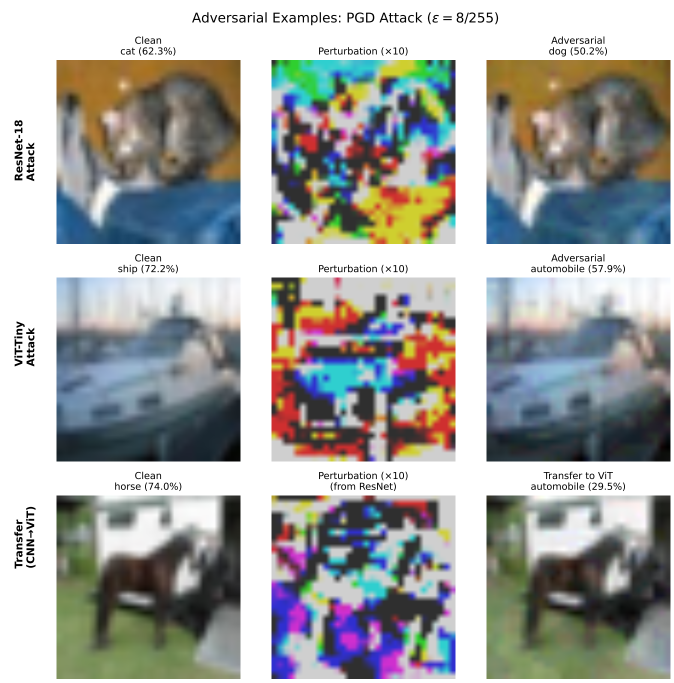



Fig. 3: For each model, its own white-box PGD-10 attack on the same test image, showing the clean prediction, the perturbation magnified 10×, and the adversarial prediction. Perturbations are bounded by $\epsilon$=8/255 and remain visually subtle at this scale.

three seed pairs. Because outcomes are logged per sample, the same attack run can be scored under several conditioning protocols without recomputation: the protocols differ only in which samples enter the denominator (Section III-F).

The denominators differ sharply between protocols. For CNN→ViT and ViT→CNN respectively they are 10,000/10,000 without conditioning, 7,353/8,579 under target-

TABLE III: Transfer asymmetry under the unconditioned rate and three established conditioning protocols (PGD-10, $\epsilon$=8/255, full test set; mean$\pm$std over three seed pairs). The rightmost column repeats the difference measured on our earlier single-run checkpoints.

| **Protocol** | **CNN→ViT** | **ViT→CNN** | **Diff.** | **Diff.**‡ |
|---|---|---|---|---|
| Unconditioned (raw) | 41.02±0.55 | 27.45±0.27 | +13.57±0.33 | +8.27 |
| Target-correct | 19.87±0.18 | 15.51±0.59 | +4.36±0.44 | +0.63 |
| Both-correct | 18.25±0.17 | 9.98±0.31 | +8.27±0.23 | +5.33 |
| Successful-source | 38.50±0.75 | 23.91±0.80 | +14.60±1.48 | +5.28 |

‡Single run, validation split seen during clean pre-training.

correct, 7,061 shared samples under both-correct, and 3,122/5,331 under successful-source. The spread quoted in the Diff. column is the standard deviation of the paired within-seed difference, which cannot be recovered from the two per-direction spreads because the directions covary across seeds, and it is the training-run reference used in Section IV-I. The result is that the protocol, not the model, dominates the measured quantity. On identical models and identical data, the asymmetry ranges from $+4.36\pm0.44$ to $+14.60\pm1.48$ points; the mean per-seed spread between the largest and the smallest protocol estimate is $10.45 \pm 0.76$ points, a factor of 3.3, whereas the seed-level standard deviation within any single protocol stays below 1.5 points. The protocol also decides the statistical verdict. Under the both-correct protocol, which scores both directions on the same samples and therefore supports paired inference, the difference is $8.27\pm0.24$ points, differing in the third decimal from the $\pm0.23$ of the table only because the two estimators differ, with a paired bootstrap CI of $[7.33, 9.21]$, a sign-flip permutation $p < 5 \times 10^{-5}$ in every seed, and TOST equivalence rejected at every margin we tested; under the target-correct protocol on our earlier checkpoints, the same pipeline accepted equivalence at $\pm2$ points. A reader given only one of these numbers would draw a different conclusion about whether the architectures differ at all.

What is stable is the direction. CNN-crafted examples transfer better to the ViT than the reverse under all four protocols, in all three seeds, and also under the momentum-based MI-FGSM [40] ($19.07 \pm 0.33$ versus $15.42 \pm 0.52\%$ target-correct), so the effect is not an artifact of the PGD attack itself. The magnitude, its significance, and any equivalence claim are protocol-dependent; the sign is not.

The mechanism behind the spread is compositional. Samples that the target solves on clean input but the source does not are three to four times more fragile than jointly solved ones ($41.2 \pm 0.8\%$ and $59.1 \pm 1.7\%$ fooling versus 10.0 and $18.3\%$), and these marginal subsets differ five-fold in size ($1{,}517 \pm 102$ versus $291 \pm 16$) because the two models differ in clean accuracy. Conditioning on target-correct admits both marginal subsets, whose asymmetric sizes pull the two rates toward each other; conditioning on both-correct excludes them, and the directional difference reappears. This account covers three of the four protocols. The successful-source protocol is driven differently, and separating the two drivers changes what the spread depends on: across 18 direction pairs drawn from six seed-and-dataset clusters, the spread of the four-protocol asymmetry *falls* as the clean-error gap between the two models widens (slope $-0.567$, cluster-bootstrap CI $[-0.757, -0.451]$), whereas the spread computed without the successful-source protocol *rises* with it (slope $+0.431$, CI $[+0.333, +0.633]$). The successful-source rate is the extreme of the four in 12 of 18 pairs, and the widest protocol pair is always target-correct against successful-source (19.68 points on average). Conditioning on source success selects by the *source's* own robustness, which is not a function of the clean-accuracy gap between the pair, so it contributes a second and partly opposing term.

The two arms agree on three protocols. We calibrate the protocol spread against the clean-error gap in two ways that fail differently. The *controlled* arm holds one member of a pair fixed and sweeps the other along its own training trajectory, so architecture, seed, dataset, and attack budget stay constant while the gap moves (116 points from eight seed-and-dataset clusters). The *observational* arm compares final models across architectures, seeds, and datasets, where everything moves at once (18 direction pairs from six clusters). Inference in both is by cluster bootstrap at the trajectory level; the degrees of freedom are set by the number of clusters, not by the number of points, and we do not pool the arms.

Excluding the successful-source protocol, the two arms give slopes of $+0.672$ (CI $[+0.601, +0.726]$) and $+0.431$ (CI $[+0.333, +0.633]$): same sign, overlapping intervals. Including it, the controlled arm gives $+0.293$ (CI $[+0.206, +0.489]$) while the observational arm gives $-0.567$ (CI $[-0.757, -0.451]$). That disagreement is a composition effect, not a control effect: the controlled arm contains no WideResNet, and restricting the observational arm to the same architecture pair yields $+0.387$ (CI $[+0.075, +0.576]$), which agrees with the controlled arm in both sign and interval. Two limits belong with this: the observational arm's negative slope is fitted across an unmeasured gap of 7.2 points on the $x$ axis, so it compares two separated clusters rather than tracing a trend; and the controlled arm's slope survives dropping early checkpoints ($+0.220$ to $+0.524$ across thresholds), with the three-protocol slope the most stable quantity of all (0.659 to 0.672).

*1) How much of the raw rate is the target's clean error?:* To quantify the confound directly rather than argue it, we extend the analysis to a $3\times3$ matrix by adding the WideResNet-28-10 reference as both source and target (Table IV). The relationship between the two rates turns out to be stronger than a correlation: it is an identity resting on a single empirical fact.

Write $e$ for the target's clean error, $r_{\text{raw}}$ for the unconditioned rate and $r_{\text{cond}}$ for the target-correct conditioned rate. If a sample the target already misclassifies on clean input is still misclassified after the perturbation, then $r_{\text{raw}} = e + r_{\text{cond}}(1-e)$, and therefore

$$r_{\text{raw}} - r_{\text{cond}} = e\,(1 - r_{\text{cond}}). \tag{11}$$

The antecedent is the part that is measured, and it holds tightly: across 36 off-diagonal directions (two datasets, three seed pairs, three targets, two sources) the probability that a clean-misclassified sample is also misclassified under attack

TABLE IV: 3 × 3 transfer matrix (PGD-10, $\epsilon$=8/255, full test set; mean±std over three seed pairs; the WideResNet column and row use the single external checkpoint). Raw = unconditioned fooling rate, Cond. = target-correct conditioned rate. Diagonal entries are white-box attacks.

| Source | Target | Raw (%) | Cond. (%) |
|---|---|---|---|
| ResNet-18 AT | ResNet-18 AT | 55.84±0.53 | 48.52±0.84 |
| ResNet-18 AT | ViT-Tiny AT | 41.07±0.58 | 19.93±0.20 |
| ResNet-18 AT | WRN-28-10 | 21.69±0.09 | 12.48±0.10 |
| ViT-Tiny AT | ResNet-18 AT | 27.45±0.34 | 15.52±0.64 |
| ViT-Tiny AT | ViT-Tiny AT | 67.34±0.18 | 55.57±0.44 |
| ViT-Tiny AT | WRN-28-10 | 18.25±0.12 | 8.65±0.14 |
| WRN-28-10 | ResNet-18 AT | 32.63±0.14 | 21.48±0.31 |
| WRN-28-10 | ViT-Tiny AT | 39.73±0.25 | 18.15±0.28 |
| WRN-28-10 | WRN-28-10 | 33.03 | 25.16 |

ranges from 0.989 to 1.000. Equation (11) consequently reproduces every measured unconditioned rate to within 0.41 points, with a mean absolute residual of 0.095 points; the residual accounts for at most 1.39% of the deviation being explained.

This changes the status of the claim, and strengthens it. The deviation of unconditioned from conditioned transfer rates does not merely correlate with the target's clean error in our data; it is that error, scaled by $(1-r_{\text{cond}})$, up to a residual below half a point. Unconditioned transfer rates are contaminated by target weakness *by construction*, in any study that reports them, not as a regularity we happened to observe on three targets. We previously stated this result as a Pearson correlation ($r = 0.997$ over six directions drawn from three targets) and that phrasing understated it: a near-unit correlation between a quantity and its own dominant component is arithmetic, and the small-sample caveat it required is unnecessary once the identity is written down.

What remains dataset-specific is the proportionality factor, not the mechanism. Because $(1 - r_{\text{cond}})$ depends on how transferable the attack is, the fitted slope is 0.762 points per point of target error on CIFAR-10 and 0.656 on CIFAR-100. It is an average over the targets measured rather than a constant, and it should be re-estimated, not assumed, when the identity is applied to other model pools. Unconditioned transfer rates therefore do not measure transferability across architectures; they measure transferability plus the target's clean weakness, in a proportion that Equation (11) makes explicit.

The third architecture also reframes the asymmetry itself. Averaged over the two foreign sources, incoming conditioned transfer is 18.50% for the ResNet target, 19.04% for the ViT target, and 10.57% for the WideResNet target, which tracks each target's own white-box conditioned fooling rate (48.52, 55.57 and 25.16%) with $r = 0.986$ over the three targets. On this evidence the CNN→ViT advantage is better read as a property of the weaker target than as evidence that CNN-crafted perturbations are intrinsically more transferable: the most robust target absorbs attacks from both families almost equally well. With only three targets this correlation is suggestive rather than conclusive, and we report it as such.

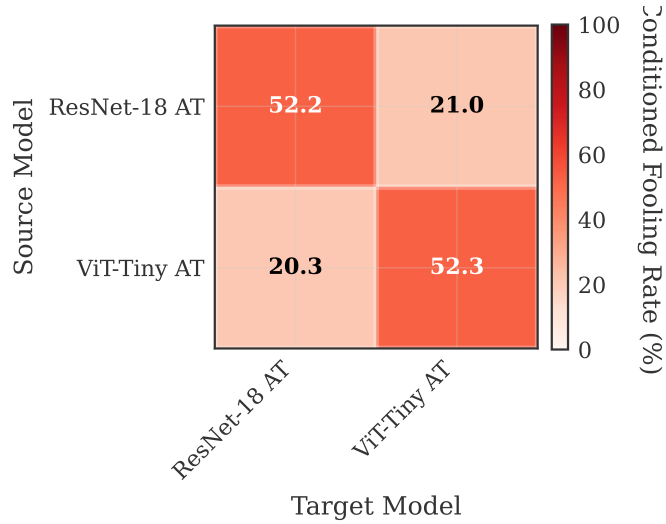



Fig. 4: Conditioned (target-correct) transfer fooling rates, mean over three seed pairs. Off-diagonal asymmetry under this protocol is +4.4 points; Table III shows how far the same quantity moves under other protocols.

TABLE V: CIFAR-100 replication (mean±std over three independent training runs per architecture, full test set, $\epsilon$=8/255). Both models are adversarially trained with the protocol of Section III-E. Absolute robustness is lower than on CIFAR-10 for both architectures, and the CNN-over-ViT ordering is unchanged.

| Model | Clean (%) | PGD-10 (%) | AutoAttack (%) |
|---|---|---|---|
| ResNet-18 AT | 63.86±1.05 | 19.30±0.32 | 15.04±0.54 |
| ViT-Tiny AT | 43.17±1.10 | 11.15±0.60 | 8.87±0.83 |

### *C. Does the Protocol Effect Generalize? A Second Dataset*

The measurement above is made on one dataset, and a protocol effect that exists only on CIFAR-10 would not support a general reporting recommendation. We therefore repeated the whole design on CIFAR-100: same architectures, same threat model, same attack budget, same three-seed structure, same four protocols. Before running it we registered three directional predictions and three endpoint bands in an append-only document, so that the outcome could not be reinterpreted after the fact.

The protocol spread grows rather than shrinks. Across the same four protocols the measured CNN→ViT asymmetry runs from $+4.96 \pm 1.01$ points (target-correct) to $+18.53 \pm 0.71$ points (unconditioned), with $+10.92 \pm 0.34$ under both-correct and $+11.44 \pm 1.82$ under successful-source. The mean per-seed spread is $13.58 \pm 1.71$ points, against $10.45 \pm 0.76$ on CIFAR-10. The direction is positive in all twelve measurements (four protocols × three seed pairs), as it was on CIFAR-10. Under the both-correct protocol the paired difference is 10.92 points with a bootstrap CI of $[9.48, 12.36]$, a sign-flip permutation $p$ below the resolution of 20,000 permutations in every seed, and equivalence rejected at every margin tested.

Two of the three registered predictions are confirmed, and the third is not testable in this design. The prediction that the protocol spread would be larger on the harder dataset is confirmed ($13.58 > 10.45$), as is the prediction that the sign would be preserved ($12/12$). The third predicted that the deviation of the unconditioned from the conditioned rate would grow with the target's clean error. The measured slope is $+0.656$ points per point of target error, positive as predicted; but on CIFAR-100 two of the three targets have clean errors that differ by $0.21$ points ($36.15$ and $36.36$), so the correlation is computed over two effective clusters rather than three, and its value ($r = 0.931$ over six directions, $0.974$ aggregated to targets) rests on a two-point contrast. The measurement is *consistent with* the prediction and does not *test* it; we report it as such rather than as a third confirmation.

One registered endpoint was missed. We predicted clean accuracy in the $33$ to $40\%$ band for the adversarially trained ViT and measured $43.17 \pm 1.10\%$. The band was derived by assuming that the clean-accuracy cost of adversarial training scales the same way for both architectures. It does not: the ViT retained $0.885$ of its clean pre-training accuracy under adversarial training, against $0.802$ for the ResNet. The two adversarial endpoints, which are the ones the protocol claim depends on, fell inside their bands (PGD-10 predicted $10$ to $14\%$, measured $11.15$; AutoAttack predicted $8$ to $11\%$, measured $8.87$).

Class composition does not manufacture the asymmetry. Because conditioning changes which samples enter the denominator, it also changes the class mix, and an asymmetry could in principle be a composition artifact. We decompose each direction's rate into a composition component and a rate component. Under target-correct the composition effect is $-1.249$, $-1.330$ and $-1.065$ points across the three seed pairs, that is $12.9$ to $18.6\%$ of the asymmetry and *negative*: rebalancing the class mix would make the asymmetry larger, not smaller. Under successful-source it is negligible and its sign is not stable ($+0.039$, $+0.013$, $-0.174$). Under the unconditioned and both-correct protocols it is exactly zero, which is the arithmetic check the decomposition must pass.

Two qualifications belong with this result. First, the two adversarial bands held with margins of $0.22$ points (AutoAttack) and $0.48$ points (PGD-10) at the nearest seed-level edge. Those margins are smaller than the checkpoint-selection noise on this same dataset: sweeping early-stopping patience and curve smoothing over the CIFAR-100 trajectories, with the training run held fixed, moves the reported PGD-10 accuracy by $0.18$ to $1.83$ points (mean $0.86$). The same nine-cell sweep on CIFAR-10 gives $0.54$ to $2.83$ points per validation split, and the eighteen-cell figure quoted in Section IV-I is larger still because it also varies the split, a dimension the CIFAR-100 protocol does not have. The bands held, but by margins at or below the selection noise of the dataset they were registered for; a reader should weigh the pre-registration accordingly. Second, the validation split used to select the adversarially trained checkpoint is the one already used during clean pre-training. That choice is inherited deliberately from the CIFAR-10 protocol so the two datasets remain comparable, and Section IV-I reports what it does and does not cost.

TABLE VI: SVHN transfer asymmetry under the four protocols (PGD-10, $\epsilon$=8/255, full test set $n$=26,032; mean±std over two seed pairs; two architectures only, so no third-architecture control). Unlike CIFAR-10 and CIFAR-100, the sign is not the same in all four rows.

| **Protocol** | **CNN→ViT** | **ViT→CNN** | **Diff.** |
|---|---|---|---|
| Unconditioned (raw) | 31.89 | 31.49 | +0.39±0.08 |
| Target-correct | 26.32 | 27.33 | −1.00±0.22 |
| Both-correct | 25.36 | 25.70 | −0.35±0.01 |
| Successful-source | 62.50 | 59.86 | +2.64±0.03 |

### D. A Third Dataset Where the Architectures Are Matched

CIFAR-10 and CIFAR-100 both place the two architectures far apart in clean accuracy (about 11 and 21 points). SVHN does not: adversarially trained under the same recipe, the ResNet reaches $94.27 \pm 0.70\%$ clean and the ViT $92.42 \pm 0.56\%$, a gap of $1.85$ points. Robust accuracies are $55.52 \pm 0.64$ and $52.30 \pm 0.10$ under PGD-10. This is the regime our thesis makes the sharpest prediction about, and the prediction is not the one a reader of the previous sections would expect.

When the architectures are matched, the protocol decides the sign. Two of the four protocols report a CNN advantage and two report a ViT advantage (Table VI). Under the both-correct protocol, which we treat as primary because it scores both directions on identical samples, the difference is $-0.35$ points with a paired bootstrap CI of $[-0.77, +0.06]$, a sign-flip permutation $p$ of $0.105$, and TOST equivalence *accepted*. A study reporting only the successful-source rate would announce a 2.64-point CNN advantage on SVHN; one reporting only the target-correct rate would announce a 1.00-point ViT advantage. Both would be arithmetically correct and neither would be reproducible by the other.

This behaviour is consistent with the mechanism rather than an exception to it. The protocol spread on SVHN is $3.65 \pm 0.19$ points, against $10.45 \pm 0.76$ on CIFAR-10 and $13.58 \pm 1.71$ on CIFAR-100, ordered exactly as the clean-accuracy gaps are ($1.85$, $\approx 11$, $\approx 21$ points). A smaller gap produces a smaller spread; and once the spread is small enough, it no longer clears the asymmetry itself, so the sign becomes a property of the protocol rather than of the architectures. The direction we reported as stable in Section IV-B is stable *in the regime where the two models differ in clean accuracy*, which is the regime both CIFAR datasets happen to occupy.

SVHN was run in a reduced configuration fixed in advance: two seed pairs rather than three, 50 adversarial-training epochs, no AutoAttack, and no third architecture. The last of these means the $3 \times 3$ confound control of Section IV-B1 cannot be built here, and we do not build a within-SVHN version of it.

### E. Does the Protocol Effect Survive a Change of Threat Model?

Everything so far is measured inside one threat model. To test whether the protocol effect is an $L_\infty$ artifact, we re-evaluated the same CIFAR-10 checkpoints under an $L_2$ budget of $\epsilon = 0.5$, registering three directional predictions before looking at any $L_2$ output. The models are $L_\infty$-trained, so

TABLE VII: Same checkpoints, $L_2$ budget ($\epsilon$=0.5; mean±std over three seed pairs). PGD-L2 uses the full test set; AutoAttack-L2 uses a pre-registered 5,000-sample subset, so its clean column differs slightly from the full-set value and the two are not drawn from the same samples.

| Model | Clean | PGD-L2 | Clean (5k) | AA-L2 |
|---|---|---|---|---|
| ResNet-18 AT | 85.78 | 63.78±0.38 | 85.33 | 60.61±0.42 |
| ViT-Tiny AT | 73.53 | 51.72±0.23 | 73.61 | 49.85±0.20 |

this measures the *measurement axis*, not $L_2$ robustness, and the absolute values below are not comparable with $L_2$-trained references.

All three registered predictions hold. The protocol spread is 10.91 points under $L_2$ against 10.45 under $L_\infty$, which is not merely non-zero but the same size. The sign is positive in all twelve measurements, and the deviation of unconditioned from conditioned rates still rises with the target's clean error (slope $+0.914$, $r = 0.999$ over eighteen directions). The four protocols report asymmetries of $+13.03 \pm 0.81$ (unconditioned), $+2.12 \pm 0.18$ (target-correct), $+4.07 \pm 0.36$ (both-correct) and $+9.51 \pm 0.89$ (successful-source). Absolute robustness keeps the CNN-over-ViT ordering under both attacks, with McNemar $p < 10^{-57}$ in every seed pair.

The slope also confirms the identity out of sample. Equation (11) predicts a slope of $1 - r_{\text{cond}}$ averaged over targets. Conditioned rates are far lower under this $L_2$ budget than under $L_\infty$ (roughly 2 to 9% against 9 to 22%), so the identity predicts a *steeper* slope, and the measured 0.914 sits where it should, against 0.762 on the same models under $L_\infty$. The proportionality factor is a property of the attack budget, not a constant, exactly as Section IV-B1 states.

### F. Gradient Characteristics

To characterize architectural differences in input sensitivity, we analyze the gradient properties of both architectures, together with a CIFAR-native ViT control (32×32 inputs, 4×4 patches, no upsampling) trained with the same adversarial protocol, which isolates the effect of the 32→224 upsampling in our main ViT pipeline. Table VIII summarizes the key statistics using scale-invariant sparsity measures (Section III-F); gradients are computed with per-sample losses so that norms do not depend on the evaluation batch size. Alignment there is the mean absolute cosine similarity over all $500 \times 499/2$ sample pairs, and every ResNet to ViT difference on the scale-invariant sparsity metrics is significant under per-sample paired Wilcoxon tests in every run, with Holm-corrected $p < 10^{-11}$ and paired Cohen's $d$ between 0.32 and 1.23.

Across all three scale-invariant sparsity measures, CNN gradients are sparser than ViT gradients, and the ordering reproduces in every run with seed-level standard deviations at least three times smaller than the gap. Because the same 500 samples are measured under both models, the differences are tested with per-sample paired statistics: all three are significant in all three runs (Wilcoxon signed-rank, Holm-corrected $p < 10^{-11}$ throughout) with paired Cohen's $d$ between 0.32 and 1.23. The ordering is a product of adversarial training rather

TABLE VIII: Gradient characteristics computed from per-sample loss gradients ($n$=500 test samples per run; mean±std over three runs).

| Metric | ResNet-18 AT | ViT-Tiny AT | ViT (native)‡ |
|---|---|---|---|
| Hoyer sparsity | **0.4928**±0.0120 | 0.4561±0.0055 | 0.405 |
| Gini coefficient | **0.6498**±0.0089 | 0.6099±0.0051 | 0.571 |
| Frac. <1% of max | **34.8%**±2.1 | 26.8%±0.9 | 17.9% |
| Alignment (absolute) | 0.0378±0.0009 | **0.0562**±0.0004 | 0.079 |
| Alignment (signed) | 0.0014 | 0.0005 | – |

‡CIFAR-native ViT control (4×4 patches, no upsampling; 5.4M params), a single checkpoint included only to test the upsampling confound.

TABLE IX: Spatial locality of input gradients ($n$=500 samples per run; mean±std over three runs). Paired differences are computed on the same samples; $p$ is the largest per-run paired Wilcoxon $p$-value. Lower area and entropy mean more localized; higher Moran's $I$ means more clustered.

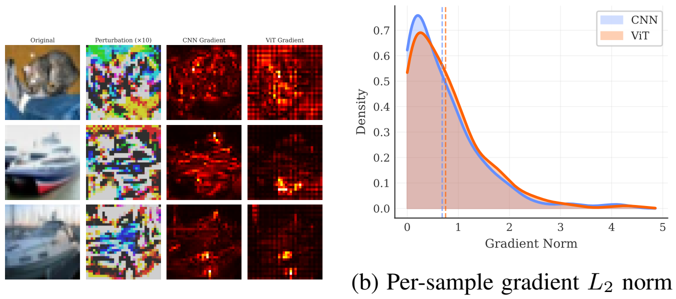
| Measure | ResNet-18 | ViT-Tiny | Diff. | $p$ |
|---|---|---|---|---|
| Area for 50% energy | 0.0378±0.0017 | 0.0397±0.0010 | −0.0019 | 0.26 |
| Area for 90% energy | 0.2345±0.0078 | 0.2570±0.0067 | −0.0224 | 0.039 |
| Spatial entropy (nat) | 5.2014±0.0387 | 5.2524±0.0197 | −0.0509 | 0.80 |
| Moran's $I$ | 0.4141±0.0132 | 0.3941±0.0024 | +0.0200 | 0.54 |

than of architecture: in the clean-trained counterparts of the same runs the *ViT* is the sparser model (Hoyer $0.380 \pm 0.009$ versus $0.347 \pm 0.002$), so adversarial training sparsifies CNN gradients far more strongly, consistent with [27]. The native-resolution ViT control is less sparse than the upsampled ViT on every measure, so the ordering is not an artifact of the $32 \to 224$ upsampling. Cross-sample alignment is higher for the ViT in absolute cosine, and the clean-trained pair already shows this gap (0.058 versus 0.029), which makes it a property of the architecture rather than of the defense. The *signed* mean cosine is near zero for both models ($\leq 0.0014$): what samples share is sensitivity axes, not consistent directions, so the alignment gap is not evidence for cheap universal perturbations.

These gradient results are also the one part of our analysis that is insensitive to which checkpoint set is used: every quantity in Table VIII reproduces the value measured on our earlier single-run checkpoints to within the reported standard deviations, in contrast to the conditional decomposition of Section IV-A.

*1) Sparsity is not spatial locality:* Sparsity measures answer how *many* gradient components are large, not whether those components sit together in the image. The distinction matters because the sparser-CNN result is frequently described in spatial language, with gradients said to be more "localized" or "concentrated" on object regions, and that reading implies a claim these measures cannot support. We therefore measure spatial structure directly on the same gradients, using four scale-invariant descriptors of the channel-summed energy map: the smallest pixel fraction carrying 50% and 90% of the gradient energy, the spatial entropy of the normalized energy map, and Moran's $I$ with 4-neighbourhood weights, which is positive when energy clusters in adjacent pixels.

The result is negative (Table IX). Three of the four measures




(a) Gradient field visualization

(b) Per-sample gradient $L_2$ norm distribution

Fig. 5: Gradient characteristics comparison. (a) Input-gradient magnitude maps for representative test images with the corresponding PGD-10 perturbations. (b) Distribution of per-sample gradient $L_2$ norms for both architectures.

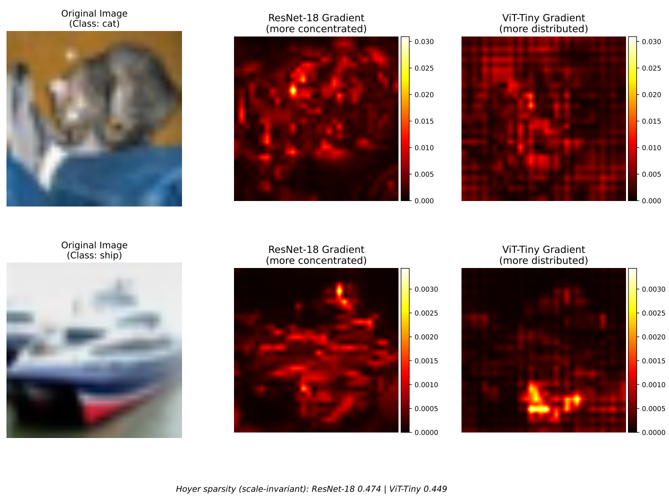



Fig. 6: Gradient heatmap comparison between CNN and ViT on the same input images. CNN gradients (top) are modestly more concentrated around object edges, while ViT gradients (bottom) are somewhat more spatially distributed (Hoyer sparsity annotations from the gradient-analysis artifacts).

show no significant difference, and the one that reaches nominal significance, the area carrying 90% of the energy, points in the direction of a *smaller* area for the CNN by two percentage points of image area, which is a marginal effect next to a Hoyer gap that is significant at $p < 10^{-11}$. Neighbourhood clustering is essentially identical for the two architectures. The two families of measures therefore disagree, and the disagreement is informative: adversarially trained CNN gradients place more of their mass on fewer pixels, but those pixels are not more spatially grouped than the ViT's. Descriptions of CNN gradients as spatially localized are not supported by our measurements, and we restrict our own claims to sparsity.

Figure 5 provides visual evidence for these gradient differences.

Figure 6 shows a direct comparison of gradient heatmaps, highlighting the structural differences in how each architecture computes input gradients.

TABLE X: Block-wise clean-to-adversarial cosine similarity under each model's own white-box PGD-10 attack ($n$=1000 per run; mean±std over three runs). ViT columns differ only in how a block's tokens are aggregated.

| | ViT-Tiny AT | | | ResNet-18 AT | |
|---|---|---|---|---|---|
| **Blk** | **CLS** | **Patch mean** | **All tokens** | **Block** | **Cosine** |
| 1 | 0.9964 | 0.9990 | 0.9975 | layer1.0 | 0.9951 |
| 2 | 0.9893 | 0.9954 | 0.9911 | layer1.1 | 0.9918 |
| 3 | 0.9785 | 0.9928 | 0.9821 | layer2.0 | 0.9879 |
| 4 | 0.9709 | 0.9877 | 0.9734 | layer2.1 | 0.9853 |
| 5 | 0.9652 | 0.9870 | 0.9719 | layer3.0 | 0.9753 |
| 6 | 0.9596 | 0.9873 | 0.9712 | layer3.1 | 0.9607 |
| 7 | 0.9497 | 0.9870 | 0.9698 | layer4.0 | **0.8777** |
| 8 | 0.9414 | **0.9840** | **0.9663** | layer4.1 | 0.9108 |
| 9 | 0.9353 | 0.9850 | 0.9675 | | |
| 10 | **0.9343** | 0.9877 | 0.9696 | | |
| 11 | 0.9352 | 0.9912 | 0.9753 | | |
| 12 | 0.9779 | 0.9955 | 0.9884 | | |

## G. *Feature Degradation in Vision Transformers*

We next measure where adversarial damage accumulates along depth, extending layer-wise representation analyses of clean-trained ViTs [28] to the adversarially trained setting. Two design choices in this measurement are usually left implicit and turn out to matter, so we vary both: the sample size ($n = 1000$ per run rather than a single small batch) and the token aggregation used to summarize a transformer block. For the ViT we report the CLS token alone, the mean over the 196 patch tokens, and the flattened concatenation of all tokens; for the CNN each residual block output is flattened over its spatial dimensions.

Two results follow. First, the two architectures differ less in *whether* damage accumulates than in *where*. In the ViT the drift is gradual and shallow: similarity declines steadily to a minimum around blocks 8 to 10 and then recovers toward the output, while feature norms stay within 0.7% of their clean values throughout, so the perturbation rotates the representation rather than rescaling it. In the CNN the damage is concentrated: the first three stages lose little, then the first block of the last residual stage collapses to $0.8777 \pm 0.0181$ together with a $-13.14 \pm 1.89\%$ norm contraction, after which the final block partially repairs the representation ($0.9108 \pm 0.0060$, norm $+5.84 \pm 3.53\%$). Adversarial damage in the CNN is thus a late, localized, magnitude-changing event; in the ViT it is a distributed, direction-only rotation. We note that on our earlier checkpoints the CNN profile fell monotonically to the last block, with no recovery, so the shape of the CNN curve, unlike the ViT profile, should be read as recipe-dependent.

Second, the aggregation choice changes the measured profile substantially. The CLS pathway drifts about three times as far as the patch-token mean (minimum 0.9343 versus 0.9840), and the two summaries even place the minimum at different depths (block 10 versus block 8). Since the CLS token is what the classifier reads, this is not a nuisance but a result: the perturbation concentrates on the classification pathway while the average patch representation is comparatively preserved. It also means that layer-wise drift curves from different papers are not directly comparable unless the aggregation is stated.

A third measurement is negative. Adversarial perturbation

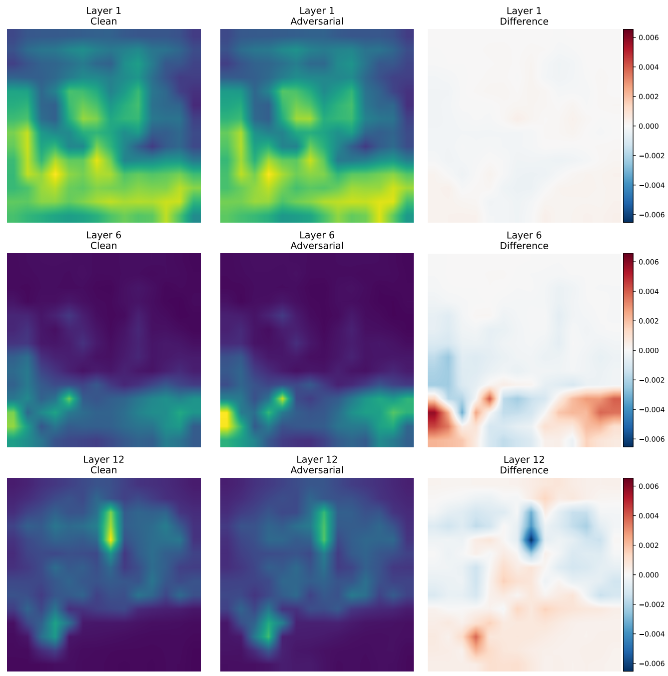
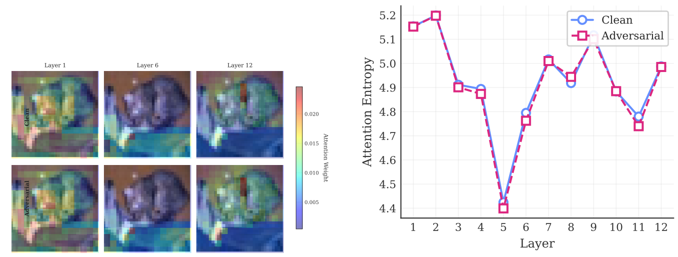



Fig. 7: CLS-token attention maps (post-softmax, head-averaged) across ViT layers under adversarial attack. Each row shows a different layer (1, 6, 12); columns show clean attention, adversarial attention, and their difference. Early layers maintain similar attention patterns, while later layers show larger attention shifts.




(a) Attention comparison (b) Layer-wise attention entropy

Fig. 8: ViT attention analysis under attack: (a) clean vs. adversarial CLS attention for a representative sample; (b) per-layer entropy of the CLS attention distribution for clean and adversarial inputs.

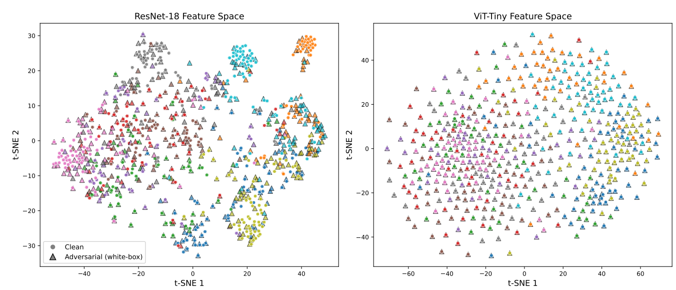



Fig. 9: Illustrative t-SNE embedding of penultimate features for clean samples (circles) and white-box PGD-10 adversarial samples (triangles), for ResNet-18 (left) and ViT-Tiny (right). t-SNE is shown for orientation only; the quantitative comparison is given in the text.

does not measurably change the entropy of the CLS attention distribution: over $n = 1000$ samples per run, the mean per-layer change is at most $0.005$ nats in absolute value against layer means near 4.9 nats, with a seed-level standard deviation of the same order. What does grow with depth is the *displacement* of the attention map, from $0.028$ total variation in the first layer to $0.084$ in the last, small in absolute terms, since total variation is bounded by 1. The attack therefore redirects attention slightly without broadening or sharpening it. We flag this explicitly because an earlier version of this analysis, computed on an 8-sample figure batch, suggested a visible attention effect that the larger measurement does not support; claims of "attention degradation" under adversarial attack should be checked at this sample size before being made.

Finally, drift magnitude does not predict attack success. Splitting the samples by whether the attack flipped the prediction, the last-block cosine similarity of flipped and unflipped ViT samples is statistically indistinguishable (0.9895 versus 0.9877); for the CNN there is a small difference in the expected direction (0.9058 versus 0.9143). Feature drift is therefore a description of how the representation moves, not a sufficient statistic for whether the prediction changes.

Complementing the quantitative measurements, Figure 7 visualizes CLS-token attention maps at layers 1, 6, and 12 for a representative sample. The visualization is consistent with the displacement measurement above, since later layers move more than early ones, but we caution that single-sample attention panels of this kind are easy to over-read: the aggregate entropy change over 1,000 samples is null, so the visible differences reflect redirection, not diffusion. Attention-shift effects under patch perturbations [31] and attacks that explicitly target the attention mechanism [30] operate in a different regime from the $L_\infty$-bounded perturbations studied here.

Figure 8 shows the clean and adversarial CLS attention for a representative sample together with the per-layer entropy curves; the two entropy curves lie on top of each other, which is the visual form of the null result reported above.

Figure 9 shows an illustrative t-SNE embedding of the penultimate feature spaces. Because t-SNE embeddings are sensitive to hyperparameters and invite qualitative over-reading, we base all claims on metrics computed in the original feature space ($n$=500 per run, each model attacked white-box, mean$\pm$std over three runs). The attack collapses class structure most strongly in the *CNN*: clean-fit 5-NN label consistency drops from $87.3 \pm 2.6\%$ to $51.3 \pm 1.8\%$, the silhouette score falls from $0.196 \pm 0.010$ to $-0.032 \pm 0.004$, the intra/inter-class distance ratio rises from $0.511 \pm 0.013$ to $0.847 \pm 0.003$, and the relative class-centroid shift is $0.238 \pm 0.012$. The ViT's penultimate space is only weakly class-separated even on clean inputs (silhouette $-0.021 \pm 0.005$; 5-NN consistency $61.1 \pm 1.4\%$, dropping to $48.5 \pm 1.4\%$), and its centroids shift far less ($0.083 \pm 0.004$). The measurable statement is

TABLE XI: ViT→CNN transfer under TGR [12] and plain MI-FGSM at a matched budget ($n$=10,000 per run; mean±std over three runs).

| Measure (%) | TGR | MI-FGSM |
|---|---|---|
| White-box on source (raw) | 51.56±0.80 | **66.43**±0.29 |
| Transfer, target-correct | 12.70±0.41 | **15.71**±0.60 |
| Transfer, both-correct | 7.93±0.42 | **10.08**±0.30 |

therefore that the CNN's well-separated clean feature space is heavily disrupted by white-box attack, whereas the ViT's class structure is weaker to begin with and moves less in these aggregate terms; this is a statement about geometry, not about robustness, since the CNN nonetheless retains the higher robust accuracy.

### *H. Does a ViT-Specific Transfer Attack Change the Picture?*

All transfer results so far use gradient attacks that treat the ViT as a generic differentiable model. A natural objection is that the ViT direction is handicapped by this choice, since attacks designed for transformer structure are reported to transfer better. We test the objection directly with Token Gradient Regularization (TGR) [12], which suppresses extreme-valued token gradients in the attention, QKV, and MLP paths during backpropagation on the source model. Our implementation is an adaptation: the original targets standardly trained ImageNet ViTs, whereas we apply the same mechanism to our CIFAR-10 ViT-Tiny wrapper and match the attack budget to the rest of the paper ($\epsilon = 8/255$, $\alpha = 2/255$, 10 steps, momentum 1.0) rather than using ImageNet defaults. As a control we generate plain MI-FGSM examples in the same run, so the comparison is paired on identical samples and identical budget.

TGR does not improve transfer in this regime; it degrades it (Table XI). The regularization behaves as designed on the source side, trading away 14.9 points of white-box strength, but the trade does not pay off at the target: conditioned transfer falls by 3.0 points under the target-correct protocol and by 2.2 under both-correct. Paired McNemar tests on the both-correct subset favor MI-FGSM decisively in every run ($p = 6.1 \times 10^{-10}$, $1.1 \times 10^{-27}$, $7.0 \times 10^{-34}$), so this is not a seed effect.

We state the conclusion narrowly. Our setting differs from the one TGR was designed for in three respects that could each matter: the targets are adversarially trained rather than standardly trained, the data are $32 \times 32$ CIFAR-10 images upsampled inside the model rather than native ImageNet, and the source is a 5.7M-parameter ViT-Tiny. The claim we can support is that TGR's transfer advantage does not carry over to this regime, not that the method fails in general. For the purposes of this paper the relevant consequence is that the CNN→ViT asymmetry reported in Section IV-B is not an artifact of using architecture-agnostic attacks: giving the ViT direction a transformer-specific attack makes it weaker, not stronger.

### *I. Sources of Variance*

Four sources of variance bear on the comparisons above. Attack-initialization stochasticity is negligible: re-running PGD-10 with three attack seeds (42, 123, 456) on a fixed 1,000-sample subset leaves clean accuracy exactly unchanged, since the loader does not shuffle, and moves robust accuracy by a standard deviation of $0.05$ points for the CNN and $0.00$ for the ViT. Training-run variance is larger but still small: across the three independent runs per architecture, the standard deviation for the adversarially trained pair is at most $0.55$ points on clean accuracy and $0.50$ on PGD-10 (Table I).

Comparing the protocol effect against those two figures directly would be an error of scale, because the protocol effect is measured on the transfer asymmetry while both figures above are measured on absolute accuracy. We therefore quantify the protocol and training-run effects on the *same* quantity, the off-diagonal asymmetry, and report both in absolute units. Within a fixed seed pair, changing only the conditioning protocol moves the asymmetry by $10.45 \pm 0.76$ points. This figure is the mean of the per-seed ranges and slightly exceeds the $10.24$-point range of the protocol means, because the extremal protocol is not the same in every seed: in the third seed pair the largest asymmetry comes from the unconditioned rate rather than from successful-source. Retraining both architectures from scratch moves that same asymmetry by a standard deviation of only $0.23$ to $1.48$ points, the value depending on which protocol it is measured under (Table III, Diff. column). A ten-point measurement effect against a sub-$1.5$-point training effect is the ordering the paper rests on.

We deliberately do not headline the ratio of these two spreads. With three runs the denominator carries two degrees of freedom, so a $\chi^2$ interval on any single ratio is very wide: for the successful-source protocol, whose run-to-run standard deviation is the largest of the four, the $95\%$ interval on the ratio runs from $0.5$ to $6.3$ and therefore includes unity. Reported as a sensitivity rather than an estimate, the ratio spans $3.3$ to $22.7$ across dispersion measures and reference protocols, and equals $20.9$ under the like-for-like standard-deviation comparison on the both-correct protocol that we treat as primary. Two cautions bound it further. The protocol with the largest asymmetry also carries the largest run-to-run standard deviation, so numerator and denominator are not independent. And because all three runs share the single validation split drawn before any training, this standard deviation excludes split-to-split variance, which makes the ratio an upper bound. The absolute comparison survives all three caveats, which is why it, and not the fold-change, carries the claim. This ordering is the quantitative form of the paper's argument: with a protocol left unstated, a reader cannot tell a real architectural difference from a measurement choice.

A fourth source is the offline checkpoint-selection protocol, which is usually left implicit. Holding a training trajectory completely fixed and varying only how its final checkpoint is chosen, that is which validation split, what early-stopping patience, and whether the validation curve is smoothed before the argmax (eighteen combinations), moves the reported PGD-10 accuracy by $2.62$ to $2.85$ points per seed for the ResNet and $1.58$ to $2.09$ points for the ViT. The weights on disk are identical across these cells; only the rule that picks one of them differs. We report this dispersion in absolute units and deliberately not as a ratio to seed-level dispersion, because that ratio depends on which of the eighteen cells is taken as the reference: swept across all eighteen it runs from $0.67$ to $3.38$ for the ResNet and $0.63$ to $7.38$ for the ViT, falling below unity for some references and inverting the comparison.

A counterweight belongs immediately next to it. On CIFAR-100 the epoch selected by our own protocol varies by 25 epochs across the three ResNet seeds, yet the resulting test PGD-10 standard deviation is $0.32$ points. The selection *path* was unstable while the selected *outcome* was not. Neither observation generalizes without the other: selection choices can move a reported number by more than a training seed does, and they can also leave it nearly untouched, and which of the two happens is a property of the trajectory's plateau rather than of the architecture. The practical consequence for reporting is the same in both cases: the selection rule is part of the result and belongs in the methods section.

**Computational Details:** The training runs and analyses reported here were conducted on an NVIDIA RTX 5090 (32GB VRAM); the earlier single-run checkpoints used for the comparison in Table II were trained on an RTX 5060 Ti (16GB). All runs used PyTorch 2.6.0 with CUDA 12.8 and deterministic cuDNN settings. AutoAttack evaluation used the standard configuration (APGD-CE, APGD-T, FAB-T, Square) on the full 10,000-sample test set in seeded 1,000-sample chunks with per-sample outcome logging. The three-seed training pipeline required 21.7 GPU-hours in total.

## V. Discussion

### A. What the Measurements Show

The central result is an ordering of dispersions. Measured on the transfer asymmetry itself, retraining both architectures from scratch moves it by a standard deviation of $0.23$ to $1.48$ points, whereas changing only the conditioning protocol moves it by $10.45 \pm 0.76$ points (Section IV-I). We state the two spreads in absolute units rather than as a fold-change, because with three runs the interval on that ratio is too wide to carry a headline. For scale on a different quantity, re-running an attack with a new random initialization moves absolute robust accuracy by about $0.05$ points and retraining moves the adversarially trained pair by at most $0.55$. A quantity that the measurement choice moves by ten points while retraining moves it by one is not a stable property of the architectures, and reporting it without the protocol communicates almost nothing. This is not an argument against conditioning, since conditioning on correctly classified samples is established practice [9], [39], [40] and remains necessary, but an argument that the choice belongs in the result, not in a footnote.

The four protocols are not interchangeable, and each has a defensible use. The unconditioned rate should not be used for architecture comparison at all: our $3 \times 3$ matrix shows its deviation from the conditioned rate is algebraically equal to the target's clean error scaled by $(1 - r_{\text{cond}})$, which means it measures target weakness in a known proportion rather than transferability. The target-correct protocol removes that confound but admits samples the source itself cannot solve, and because those marginal subsets are hyper-vulnerable and asymmetrically sized, it pulls two directions toward each other by an amount that depends on the clean-accuracy gap. The both-correct protocol scores both directions on identical samples and is the only one of the four that supports paired inference, which is why we treat it as the primary estimate. The successful-source protocol answers a different question, how well an attack that already worked carries over, and its larger value reflects that conditioning on source success selects strong perturbations.

Across four protocols, three training runs, and two attack families (PGD-10 and MI-FGSM), adversarial examples crafted on the CNN transfer better to the ViT than the reverse *on both CIFAR datasets*. On SVHN, where adversarial training under the same recipe leaves the two architectures $1.85$ points apart in clean accuracy rather than 11 or 21, that is no longer true: two protocols report a CNN advantage and two report a ViT advantage, and under the primary both-correct protocol the difference is $-0.35$ points with equivalence accepted (Section IV-D). The direction is a property of the unmatched regime, not of the architectures. What changes is how large the effect looks and whether an equivalence test accepts it. Our three-target analysis suggests a reading of this direction that does not require the CNN's perturbations to be special: incoming transfer tracks each target's own white-box vulnerability ($r = 0.986$ over three targets), and the most robust target absorbs attacks from both families almost equally. On this account the asymmetry is a statement about which model is the weaker target, which is consistent with our finding that a transformer-specific attack [12] does not rescue the ViT direction: with three targets the correlation is suggestive, and a larger model pool would be needed to establish it.

The conditional decomposition separates absolute robustness into a clean-accuracy factor and a conditional-susceptibility factor. In our corrected runs both favor the CNN, so the decomposition and the headline agree. They did not agree on our earlier checkpoints, where conditional susceptibility appeared matched under PGD-10 and slightly ViT-favorable under AutoAttack. Since the absolute ordering was identical in both checkpoint sets, a paper reporting only absolute accuracy would have seen nothing change, while a paper reporting the decomposition from a single run would have drawn a mechanistic conclusion that does not replicate. The practical lesson is that mechanism-level claims need more training runs than ranking-level claims, and the two should not be given the same evidential weight.

Sparsity is not spatial locality. Adversarially trained CNN gradients concentrate more mass on fewer components (Hoyer $0.4928$ versus $0.4561$, paired $p < 10^{-11}$), but four spatial descriptors of the same gradients show no corresponding difference in how those components are arranged in the image. Descriptions of CNN gradients as spatially localized therefore need separate evidence; sparsity measures [55] do not provide it.

Adversarial perturbation does not measurably change CLS attention entropy. At $n = 1000$ the per-layer entropy change is at most $0.005$ nats, while attention maps are displaced by a small and depth-increasing amount. Attention-shift results obtained with patch attacks [30], [31] concern a different threat model and should not be generalized to $L_\infty$-bounded perturbations without measurement.

A ViT-specific transfer attack does not improve ViT-sourced transfer here. TGR [12] costs 14.9 points of white-box strength and returns none of it at the target in our regime. We report this as a regime-specific negative result, since our targets are adversarially trained and our data are CIFAR-scale, and it is exactly the kind of claim that a single untested assumption would otherwise import into a comparison.

Two behavioral differences reproduce across independent checkpoint sets and are unaffected by the protocol question. Gradient structure differs: CNN input gradients are sparser under all three scale-invariant measures, adversarial training rather than architecture creates that ordering (the clean-trained pair has it reversed), and cross-sample alignment is higher for the ViT in absolute cosine while the signed mean is near zero for both, so that samples share sensitivity axes, not directions, which is the distinction that separates our measurement from a universal-perturbation claim [56]. Depth profiles differ: adversarial damage in the CNN is a late, localized, magnitude-changing event confined to the last residual stage, whereas in the ViT it is a distributed, direction-only rotation. That the ViT's measured profile depends on token aggregation, since the CLS pathway drifts three times as far as the patch-token mean, is itself a reporting requirement for layer-wise studies.

### B. Reporting Requirements

The measurements above imply a short and concrete set of requirements for cross-architecture robustness studies.

Studies should state the conditioning protocol and report which samples enter the denominator. Where a directional claim is made, a protocol that scores both directions on identical samples should be preferred, so that paired inference is available. Unconditioned transfer rates should not be reported as architecture comparisons. When targets differ in clean accuracy, the raw rate is a known mixture of transferability and target weakness, $r_{\text{raw}} - r_{\text{cond}} = e\,(1 - r_{\text{cond}})$, and because this is an identity rather than a fitted relation the correction can be applied to published raw rates without re-running anything.

Studies should also report the decomposition and not only the score. Absolute robustness factors exactly into clean accuracy and conditional susceptibility, and which factor carries a headline is the interesting part, which is not recoverable from the score alone. Ranking claims should be separated from mechanism claims: our rankings were stable across checkpoint sets while our mechanism attribution was not, so mechanism claims should carry more training runs. For layer-wise analyses the aggregation should be stated, since CLS-only, patch-mean, and all-token summaries of the same blocks give different drift magnitudes and different minima. Finally, attention and locality claims should be checked at a realistic sample size, because both of the null results we report would have looked positive on a small figure batch.

For practitioners choosing between the two families under matched standard adversarial training on this benchmark, the CNN gives higher absolute robust accuracy and the advantage holds under every view we measured. The WideResNet reference at 62.76% shows that recipe, capacity, and extra data jointly [48] move robustness further than the architectural choice does, which is worth weighing before investing in architecture selection as a robustness strategy.

### C. Limitations

Our study is deliberately narrow, and the narrowness bounds the conclusions in specific ways.

The protocol-sensitivity result is measured on three datasets and one model pair. On CIFAR-100 the spread is larger than on CIFAR-10 ($13.58 \pm 1.71$ against $10.45 \pm 0.76$ points) and the sign is preserved in all twelve measurements (Section IV-C), which is consistent with a mechanism that runs through the clean-accuracy difference between targets rather than through anything CIFAR-specific. That account is incomplete, and we state the limit rather than the slogan: it holds for three of the four protocols, while the successful-source protocol adds a second driver tied to the *source's* robustness that is strong enough to reverse the overall relationship (Section IV-B). What we still cannot state is the functional form of either dependence with confidence: the estimates rest on six seed-and-dataset clusters, and every small-gap pair in our pool involves the external WideResNet, so a small clean-accuracy gap and a different training recipe are not separable here. A study spanning several architecture pairs with varying clean-accuracy gaps would turn these measurements into a calibration curve.

We cannot isolate the cause of the differences between our two checkpoint sets. The corrected protocol changed the validation handling and produced a fresh training run at the same time. A data-quantity explanation is ruled out, since the corrected clean models are better despite less data, but distinguishing selection leakage from ordinary run-to-run variation would require an ablation that trains on identical data with and without a leaked selection split. We report the contrast as evidence that single-run mechanism claims are fragile, not as a measurement of leakage.

Every model in this paper is trained inside a single threat model: $L_\infty$ at $\epsilon = 8/255$. We evaluated the same checkpoints under an $L_2$ budget and the protocol spread survived at the same size (Section IV-E), but that test probes the measurement axis rather than $L_2$ robustness, and its absolute values are not comparable with $L_2$-trained references. Whether the effect holds for models *trained* under another norm is untested here.

Three runs per architecture bound seed variance but do not span the recipe space, and ViT adversarial training is known to be recipe-sensitive on small datasets [50], [51]; with diffusion-generated synthetic data ViTs have been reported to surpass WideResNets on CIFAR-10 [35], [57]. Our ViT results characterize baseline adversarial training, not the best achievable ViT robustness. The ViT-Tiny pipeline upsamples $32 \times 32$ inputs to $224 \times 224$ inside the model; our CIFAR-native control shows the gradient ordering is not an upsampling artifact, but that control differs in capacity and patch size and therefore calibrates the confound rather than providing a second matched pair. The two models also differ in parameter count (11.2M versus 5.7M), so architectural attribution of the absolute gap remains partial. Finally, we evaluate a single-budget $L_\infty$ threat model, and our TGR result is an adaptation of the original method rather than a reproduction of its published setting.

### D. Future Directions

The most direct extension is to turn the protocol-sensitivity measurement into a general one: sweep architecture pairs with controlled clean-accuracy gaps and measure how the spread across protocols scales with that gap. If the relationship is as regular as our three-target correlation suggests, it would yield a correction that lets published raw-rate asymmetries be reinterpreted rather than discarded.

Two of the extensions we previously listed as future work have since been run, and we record both what they settled and what they did not. The leakage ablation was carried out on six full-budget trajectories, and it did not detect an effect of leakage on checkpoint selection; it answers a weaker question than the one originally posed, because the treatment was applied to a single validation split, so split identity and leakage cannot be separated within that design. Its informative by-product is the selection-protocol dispersion reported in Section IV-I. The second dataset has also been run (Section IV-C). What remains open is the capacity-matched pair (for example ViT-Small against ResNet-50), which would test whether the surviving behavioral differences, the sparsity ordering and the depth profiles, are properties of the families or of these particular small models.

## VI. Conclusion

We asked how much of the disagreement in CNN and Vision Transformer robustness comparisons comes from the measurement rather than from the models, and answered it by holding the models, the data, the threat model, and the attack budget fixed while varying only how the comparison is scored.

The measurement dominates. On identical adversarially trained ResNet-18 and ViT-Tiny pairs, the cross-architecture transfer asymmetry ranges from $+4.4$ to $+14.6$ points across four established conditioning protocols, a 3.3-fold spread, whereas retraining both architectures from scratch moves that same quantity by under $1.5$ points, and the protocol decides whether an equivalence test is accepted or rejected. Unconditioned rates are not a valid basis for such a comparison at all: in a $3 \times 3$ matrix that adds a WideResNet-28-10 reference, their deviation from conditioned rates is algebraically equal to the target's clean error scaled by one minus the conditioned rate, an identity that holds to within $0.41$ points across $36$ measured directions and that applies to any published raw rate. The same design run on CIFAR-100 widens the spread to $13.58 \pm 1.71$ points rather than shrinking it, and an ablation that fixes the trajectory and varies only the checkpoint-selection rule moves reported robust accuracy by $1.58$ to $2.85$ points, so measurement choices that are usually left implicit are of the same order as the effects being reported. The direction is stable across protocols, seeds, and attack families on both CIFAR datasets, but not across datasets: on SVHN, where the same recipe leaves the two architectures only $1.85$ points apart in clean accuracy, two of the four protocols reverse the sign and the primary protocol accepts equivalence. What survives is narrower and more useful than a ranking: the direction holds while the models are unmatched, and which our three-target analysis associates with the target's own vulnerability ($r = 0.986$) rather than with any special potency of CNN-crafted perturbations.

A conditional decomposition separates absolute robustness into a clean-accuracy factor and a conditional-susceptibility factor. Both favor the CNN in our corrected runs, but the attribution between them differed on an earlier checkpoint set whose absolute ordering was identical, which shows that mechanism-level claims are more fragile than ranking-level claims and need more training runs to support.

Three claims common in this literature did not survive direct measurement: adversarially trained CNN gradients are sparser but not spatially more localized, CLS attention entropy does not change under $L_\infty$ attack at $n = 1000$, and a transformer-specific transfer attack yields no transfer gain in this regime. Two behavioral differences did survive and reproduce across independent checkpoint sets: the gradient sparsity ordering, which adversarial training creates rather than architecture, and the depth profile of adversarial damage, which is a late localized magnitude-changing event in the CNN and a distributed direction-only rotation in the ViT.

From these measurements we distilled six reporting requirements, of which the first two matter most: state the conditioning protocol and prefer paired ones, and do not present unconditioned transfer rates as architecture comparisons. Our study covers one dataset and one model pair, so the size of the protocol effect in other settings remains to be calibrated; the mechanism behind it, an unequal clean-accuracy gap between targets, is not specific to our setting. We release the full pipeline, per-sample logs, and analysis scripts so that the protocol comparison can be reproduced and extended.

## Data and Code Availability

The experiments in this study use the CIFAR-10 dataset [11], which is publicly available at https://www.cs.toronto.edu/~kriz/cifar.html. Pre-trained WideResNet-28-10 weights were obtained from the RobustBench benchmark [22], [48]. The source code for the analysis pipeline and trained model checkpoints will be made publicly available upon acceptance.

## Acknowledgment

The authors acknowledge the computational resources provided by the Department of Software Engineering, Fırat University.

## References

[1] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus, "Intriguing properties of neural networks," in *International Conference on Learning Representations*, 2014.

[2] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," in *International Conference on Learning Representations*, 2015.

[3] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations*, 2021.

[4] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[5] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770–778.

[6] Y. Bai, J. Mei, A. L. Yuille, and C. Xie, "Are transformers more robust than cnns?" in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 26 831–26 843.

[7] P. Benz, S. Ham, C. Zhang, A. Karjauv, and I. S. Kweon, "Adversarial robustness comparison of vision transformer and MLP-Mixer to CNNs," in *Proceedings of the 32nd British Machine Vision Conference (BMVC)*, 2021.

[8] K. Mahmood, R. Mahmood, and M. Van Dijk, "On the robustness of vision transformers to adversarial examples," in *IEEE/CVF International Conference on Computer Vision*, 2021, pp. 7838–7847.

[9] D. Ravikumar, S. Kodge, I. Garg, and K. Roy, "TREND: Transferability-based robust ENsemble design," *IEEE Transactions on Artificial Intelligence*, vol. 4, no. 3, pp. 534–548, 2023.

[10] M. Naseer, K. Ranasinghe, S. Khan, F. S. Khan, and F. Porikli, "On improving adversarial transferability of vision transformers," in *International Conference on Learning Representations*, 2022.

[11] A. Krizhevsky, G. Hinton *et al.*, "Learning multiple layers of features from tiny images," University of Toronto, Tech. Rep., 2009.

[12] J. Zhang, Y. Huang, W. Wu, and M. R. Lyu, "Transferable adversarial attacks on vision transformers with token gradient regularization," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 16 415–16 424.

[13] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *International Conference on Learning Representations*, 2018.

[14] N. Carlini and D. Wagner, "Towards evaluating the robustness of neural networks," in *IEEE Symposium on Security and Privacy (SP)*, 2017, pp. 39–57.

[15] F. Croce and M. Hein, "Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks," in *International Conference on Machine Learning*, 2020, pp. 2206–2216.

[16] H. Zhang, Y. Yu, J. Jiao, E. Xing, L. El Ghaoui, and M. Jordan, "Theoretically principled trade-off between robustness and accuracy," in *International Conference on Machine Learning*, 2019, pp. 7472–7482.

[17] Y. Wang, D. Zou, J. Yi, J. Bailey, X. Ma, and Q. Gu, "Improving adversarial robustness requires revisiting misclassified examples," in *International Conference on Learning Representations*, 2020.

[18] C. Guo, M. Rana, M. Cisse, and L. Van Der Maaten, "Countering adversarial images using input transformations," in *International Conference on Learning Representations*, 2018.

[19] J. Cohen, E. Rosenfeld, and Z. Kolter, "Certified adversarial robustness via randomized smoothing," in *International Conference on Machine Learning*, 2019, pp. 1310–1320.

[20] F. Tramèr, A. Kurakin, N. Papernot, I. Goodfellow, D. Boneh, and P. McDaniel, "Ensemble adversarial training: Attacks and defenses," in *International Conference on Learning Representations*, 2018.

[21] S. Lyu, S. Shaikh, F. Shpilevskiy, E. Shelhamer, and M. Lécuyer, "Adaptive randomized smoothing: Certified adversarial robustness for multi-step defences," in *Advances in Neural Information Processing Systems*, vol. 37, 2024.

[22] F. Croce, M. Andriushchenko, V. Sehwag, E. Debenedetti, N. Flammarion, M. Chiang, P. Mittal, and M. Hein, "RobustBench: a standardized adversarial robustness benchmark," in *NeurIPS Datasets and Benchmarks Track*, 2021.

[23] D. Wu, Y. Wang, S.-T. Xia, J. Bailey, and X. Ma, "Skip connections matter: On the transferability of adversarial examples generated with ResNets," in *International Conference on Learning Representations (ICLR)*, 2020.

[24] C. Xie and A. Yuille, "Intriguing properties of adversarial training at scale," in *International Conference on Learning Representations (ICLR)*, 2020.

[25] C. Xie, M. Tan, B. Gong, J. Wang, A. L. Yuille, and Q. V. Le, "Adversarial examples improve image recognition," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 819–828.

[26] D. Tsipras, S. Santurkar, L. Engstrom, A. Turner, and A. Madry, "Robustness may be at odds with accuracy," in *International Conference on Learning Representations (ICLR)*, 2019.

[27] P. Chalasani, J. Chen, A. R. Chowdhury, X. Wu, and S. Jha, "Concise explanations of neural networks using adversarial training," in *Proceedings of the 37th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 119. PMLR, 2020, pp. 1383–1391.

[28] M. Raghu, T. Unterthiner, S. Kornblith, C. Zhang, and A. Dosovitskiy, "Do vision transformers see like convolutional neural networks?" in *Advances in Neural Information Processing Systems*, vol. 34. Curran Associates, Inc., 2021, pp. 12 116–12 128.

[29] S. Bhojanapalli, A. Chakrabarti, D. Glasner, D. Li, T. Unterthiner, and A. Veit, "Understanding robustness of transformers for image classification," in *IEEE/CVF International Conference on Computer Vision*, 2021, pp. 10 231–10 241.

[30] Y. Fu, S. Zhang, S. Wu, C. Wan, and Y. Lin, "Patch-Fool: Are vision transformers always robust against adversarial perturbations?" in *International Conference on Learning Representations (ICLR)*, 2022.

[31] J. Gu, V. Tresp, and Y. Qin, "Are vision transformers robust to patch perturbations?" in *Computer Vision – ECCV 2022*, ser. Lecture Notes in Computer Science, vol. 13672. Springer, 2022, pp. 404–421.

[32] R. Shao, Z. Shi, J. Yi, P.-Y. Chen, and C.-J. Hsieh, "On the adversarial robustness of vision transformers," *Transactions on Machine Learning Research*, 2022.

[33] S. Paul and P.-Y. Chen, "Vision transformers are robust learners," in *AAAI Conference on Artificial Intelligence*, 2022, pp. 2071–2081.

[34] S. Jain and T. Dutta, "Towards understanding and improving adversarial robustness of vision transformers," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 24 736–24 745.

[35] J. Wu, Z. Song, X. Zhang, S. Xie, L. Lin, and K. Wang, "Vision transformers beat WideResNets on small scale datasets adversarial robustness," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 1, 2025, pp. 886–894.

[36] Z. Zhu, X. Wang, Z. Jin, J. Zhang, and H. Chen, "Enhancing transferable adversarial attacks on vision transformers through gradient normalization scaling and high-frequency adaptation," in *International Conference on Learning Representations*, 2024.

[37] N. Papernot, P. McDaniel, and I. Goodfellow, "Transferability in machine learning: from phenomena to black-box attacks using adversarial samples," *arXiv preprint arXiv:1605.07277*, 2016.

[38] A. Demontis, M. Melis, M. Pintor, M. Jagielski, B. Biggio, A. Oprea, C. Nita-Rotaru, and F. Roli, "Why do adversarial attacks transfer? explaining transferability of evasion and poisoning attacks," in *28th USENIX Security Symposium*, 2019, pp. 321–338.

[39] Y. Liu, X. Chen, C. Liu, and D. Song, "Delving into transferable adversarial examples and black-box attacks," in *Proceedings of the 5th International Conference on Learning Representations (ICLR)*, 2017.

[40] Y. Dong, F. Liao, T. Pang, H. Su, J. Zhu, X. Hu, and J. Li, "Boosting adversarial attacks with momentum," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 9185–9193.

[41] Z. Zhao, H. Zhang, R. Li, R. Sicre, L. Amsaleg, M. Backes, Q. Li, Q. Wang, and C. Shen, "Revisiting transferable adversarial images: Systemization, evaluation, and new insights," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 48, no. 1, pp. 765–780, 2026.

[42] Z. Wei, J. Chen, M. Goldblum, Z. Wu, T. Goldstein, and Y.-G. Jiang, "Towards transferable adversarial attacks on vision transformers," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 3, 2022, pp. 2668–2676.

[43] W. Ma, Y. Li, X. Jia, and W. Xu, "Transferable adversarial attack for both vision transformers and convolutional networks via momentum integrated gradients," in *IEEE/CVF International Conference on Computer Vision*, 2023, pp. 4630–4639.

[44] Z. Chen, C. Xu, H. Lv, S. Liu, and Y. Ji, "Understanding and improving adversarial transferability of vision transformers and convolutional neural networks," *Information Sciences*, vol. 648, p. 119474, 2023.

[45] F. Pinto, P. H. S. Torr, and P. K. Dokania, "An impartial take to the CNN vs Transformer robustness contest," in *Computer Vision – ECCV 2022*, ser. Lecture Notes in Computer Science, vol. 13673. Cham: Springer, 2022, pp. 466–480.

[46] K. Ali, M. S. Bhatti, A. Saeed, A. Athar, M. A. Al Ghamdi, S. H. Almotiri, and S. Akram, "Adversarial robustness of vision transformers versus convolutional neural networks," *IEEE Access*, vol. 12, pp. 105 281–105 293, 2024.

[47] S. Zagoruyko and N. Komodakis, "Wide residual networks," in *Proceedings of the British Machine Vision Conference (BMVC)*. BMVA Press, 2016, pp. 87.1–87.12.

[48] S. Gowal, C. Qin, J. Uesato, T. Mann, and P. Kohli, "Uncovering the limits of adversarial training against norm-bounded adversarial examples," *arXiv preprint arXiv:2010.03593*, 2020.

[49] R. Wightman, "Pytorch image models," https://github.com/rwightman/pytorch-image-models, 2019.

[50] E. Debenedetti, V. Sehwag, and P. Mittal, "A light recipe to train robust vision transformers," in *IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*, 2023, pp. 225–253.

[51] Y. Mo, D. Wu, Y. Wang, Y. Guo, and Y. Wang, "When adversarial training meets vision transformers: Recipes from training to architecture," in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 18 599–18 611.

[52] L. Rice, E. Wong, and J. Z. Kolter, "Overfitting in adversarially robust deep learning," in *International Conference on Machine Learning*, 2020, pp. 8093–8104.

[53] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," in *International Conference on Learning Representations (ICLR)*, 2017.

[54] ——, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.

[55] N. Hurley and S. Rickard, "Comparing measures of sparsity," *IEEE Transactions on Information Theory*, vol. 55, no. 10, pp. 4723–4741, Oct. 2009.

[56] S.-M. Moosavi-Dezfooli, A. Fawzi, O. Fawzi, and P. Frossard, "Universal adversarial perturbations," in *IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 1765–1773.

[57] Z. Wang, T. Pang, C. Du, M. Lin, W. Liu, and S. Yan, "Better diffusion models further improve adversarial training," in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 36 246–36 263.

PLACE PHOTO HERE

**Çağrı Şahin** received the B.Sc. degree in software engineering from Fırat University, Elazığ, Turkey. He is currently pursuing the M.Sc. degree at the Department of Software Engineering, Fırat University. His research interests include adversarial machine learning, deep learning security, and computer vision, with a focus on the robustness of neural network architectures against adversarial perturbations.

PLACE PHOTO HERE

**Özal Yıldırım** is with the Department of Software Engineering, Fırat University, Elazığ, Turkey. His research interests include deep learning, computer vision, and biomedical signal processing.